



2.4GHz PA仿真示例

以中和差分对结构为例
集成电路设计工程 -25 秋

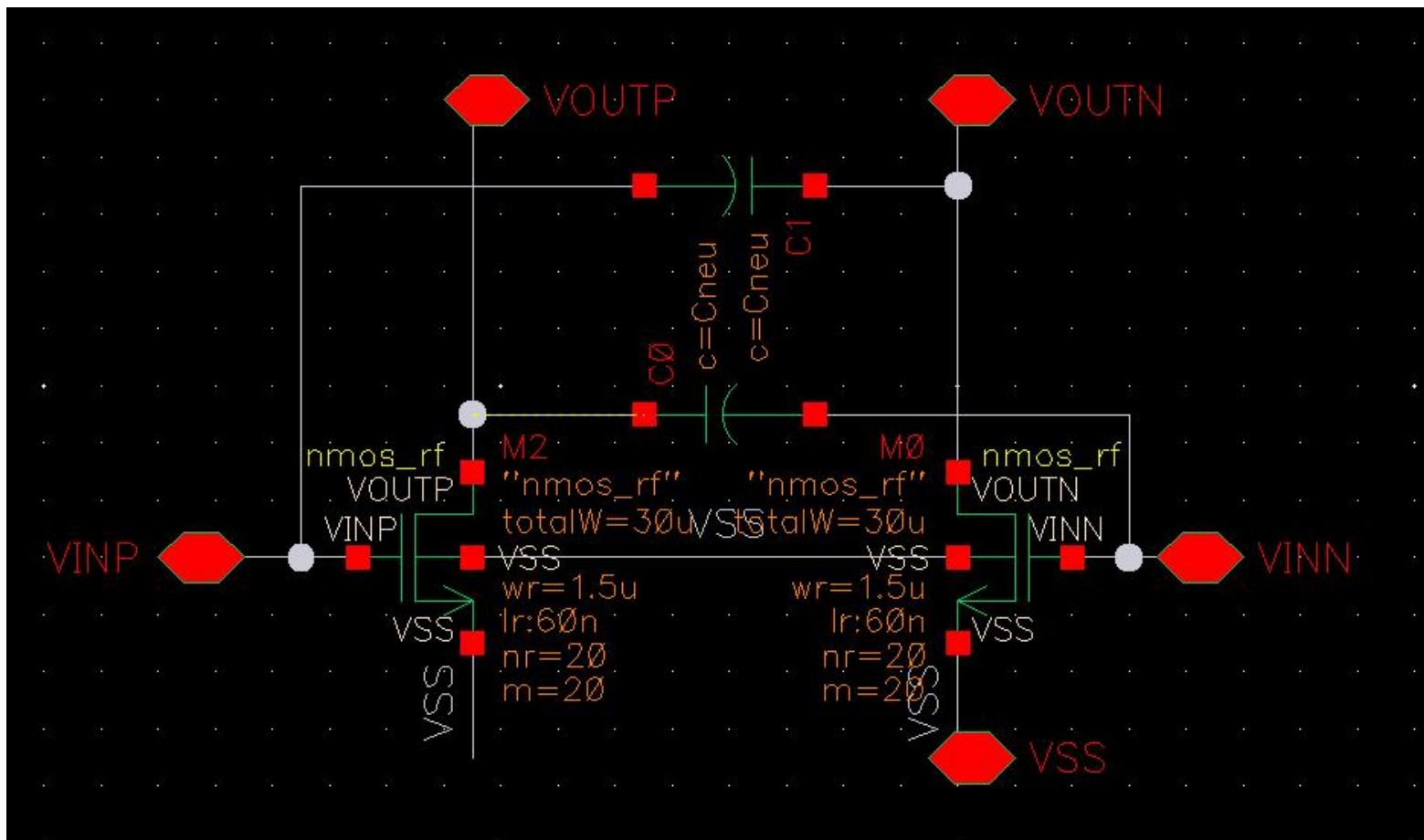


- 放大器Schematic搭建及Symbol生成
- Testbench搭建
- 中和电容选取
- 小信号(sp)仿真确认输入阻抗
- Loadpull仿真确认输出阻抗
- 核心性能获取

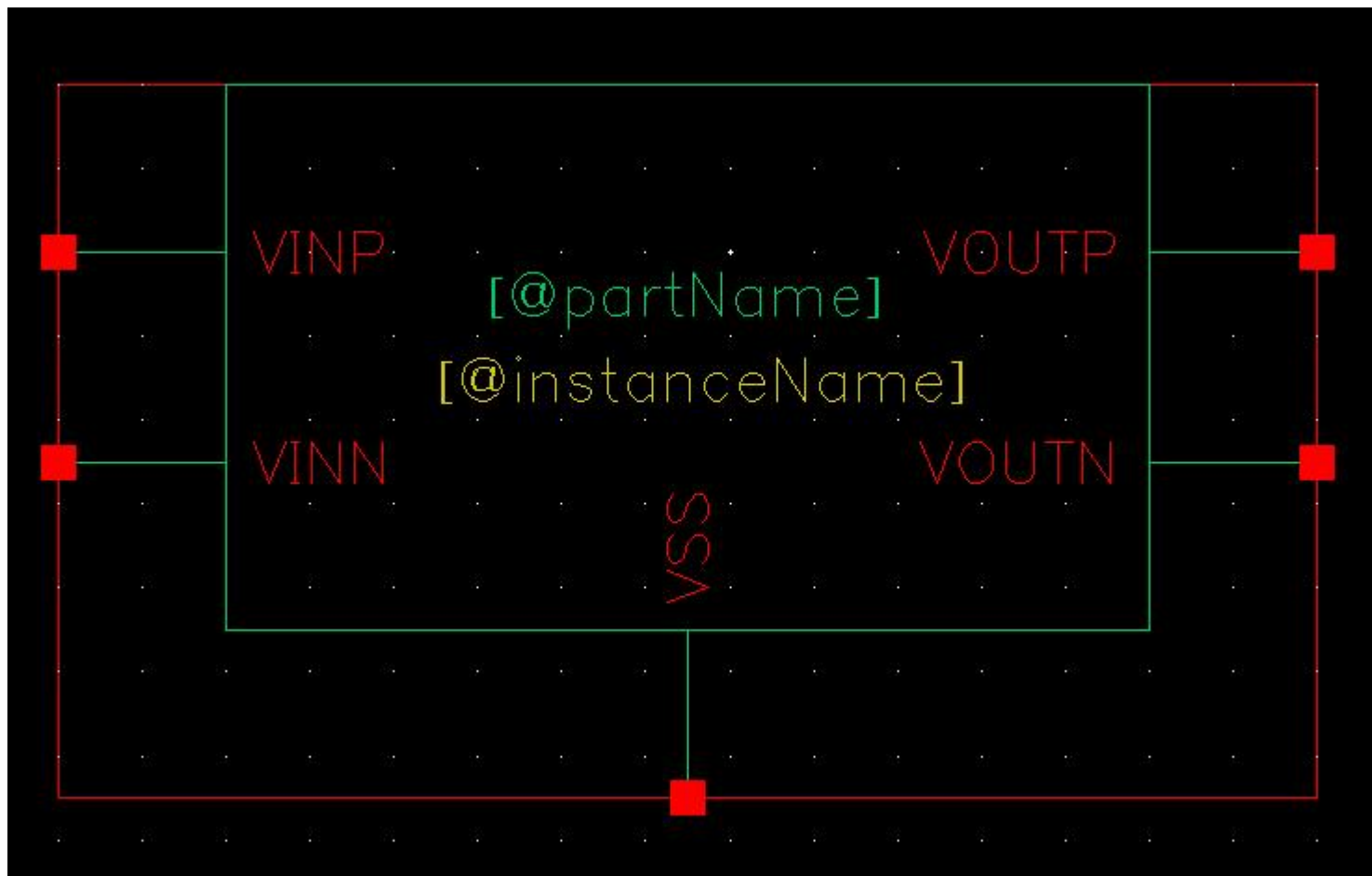


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- 复杂电路搭建时，养成**Hierarchy**的电路搭建结构。



- 将放大器schematic封装为Symbol，如下图所示.

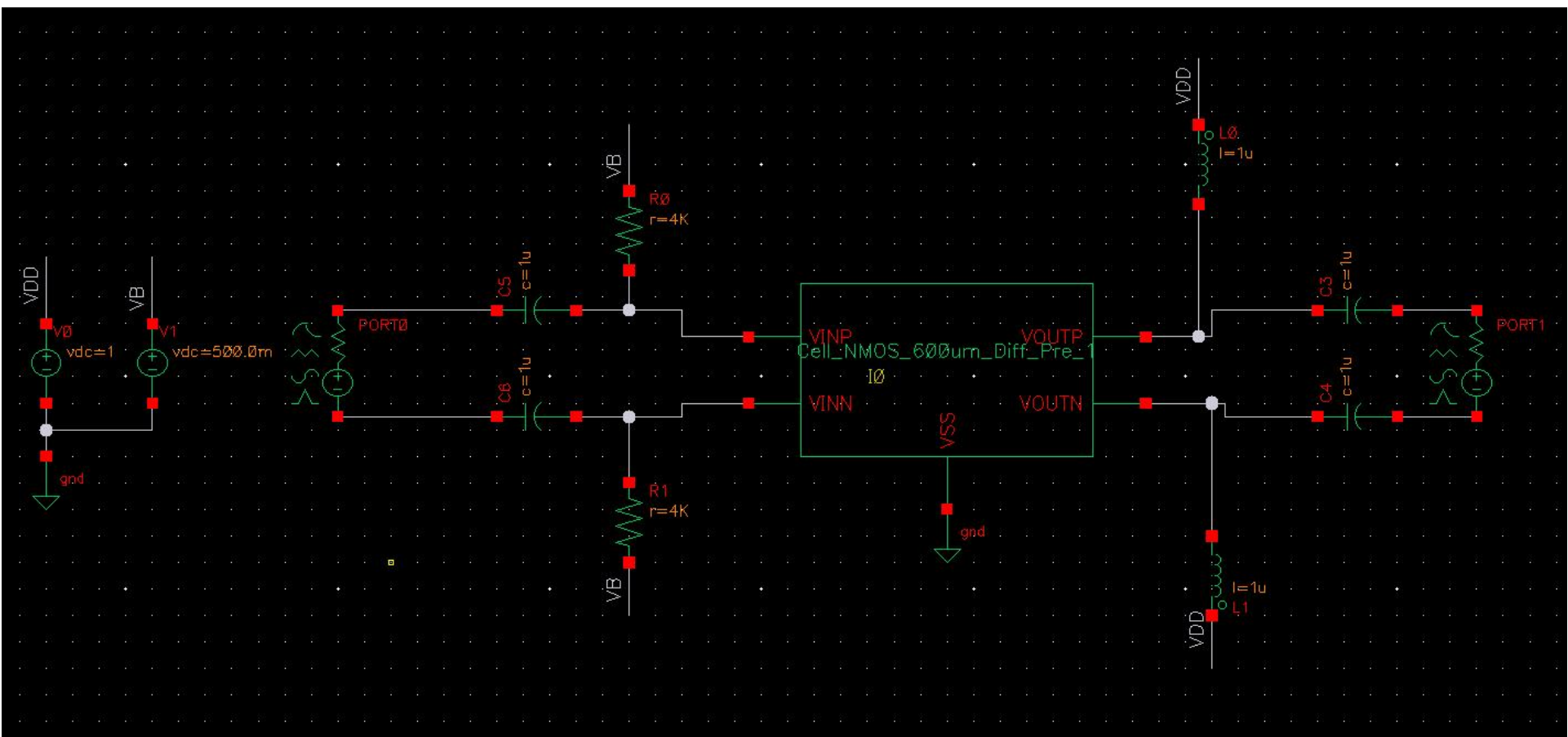




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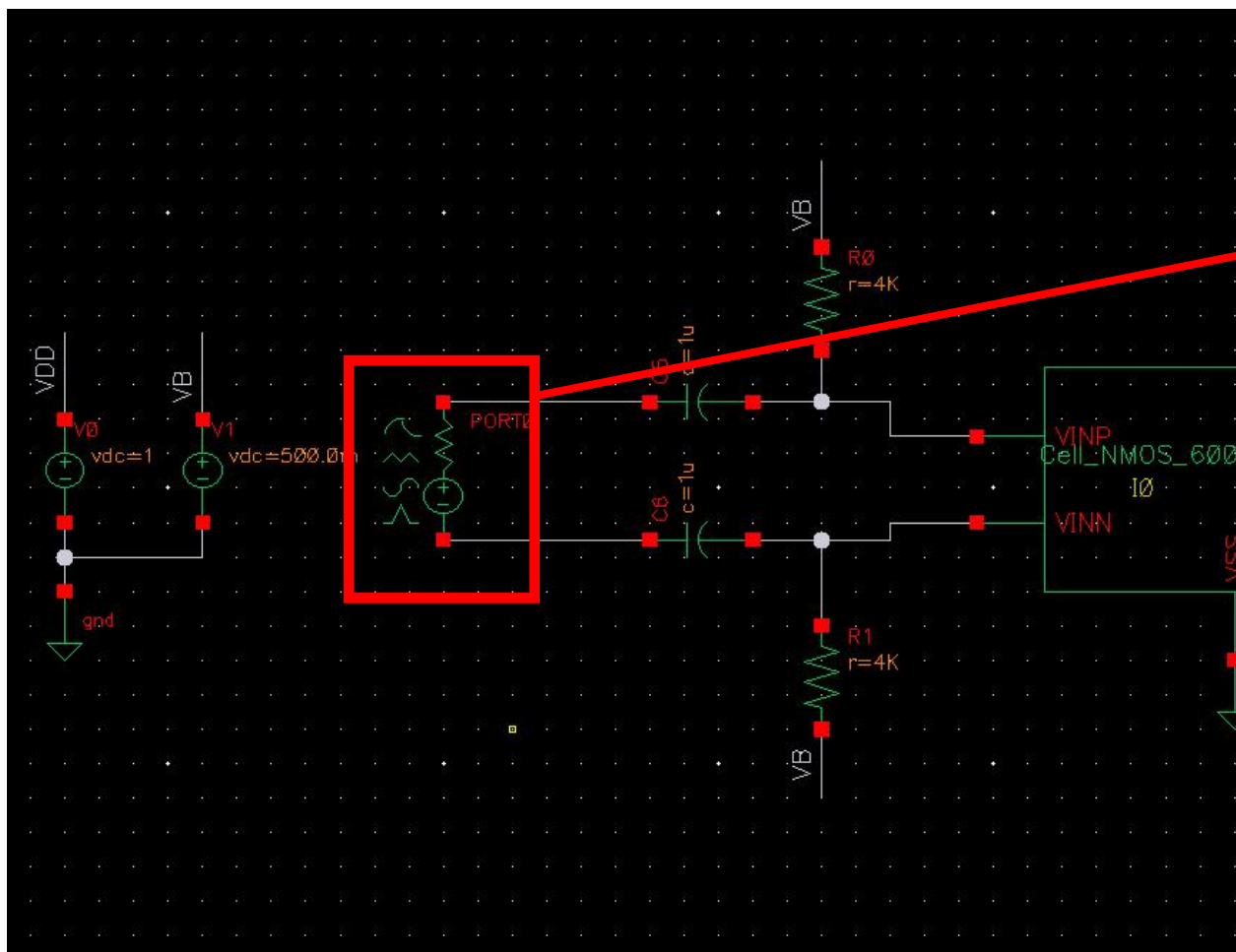
Testbench搭建

- 在相同的library下，新建cellview，命名为xxx_tb，如下图所示。
- 请注意，Testbench的电路，和cell的电路是不同的。



Testbench搭建

- Testbench中各元件的详细信息



Edit Object Properties

Apply To: ☒ only current ☒ instance

Show: ☐ system ☒ user ☒ CDF

Browse Reset Instance Labels Display

Property	Value	Display
Library Name	analogLib	off
Cell Name	port	off
View Name	symbol	off
Instance Name	PORT0	off

Add Delete Modify

User Property	Master Value	Local Value	Display
Ivignore	TRUE		off

CDF Parameter

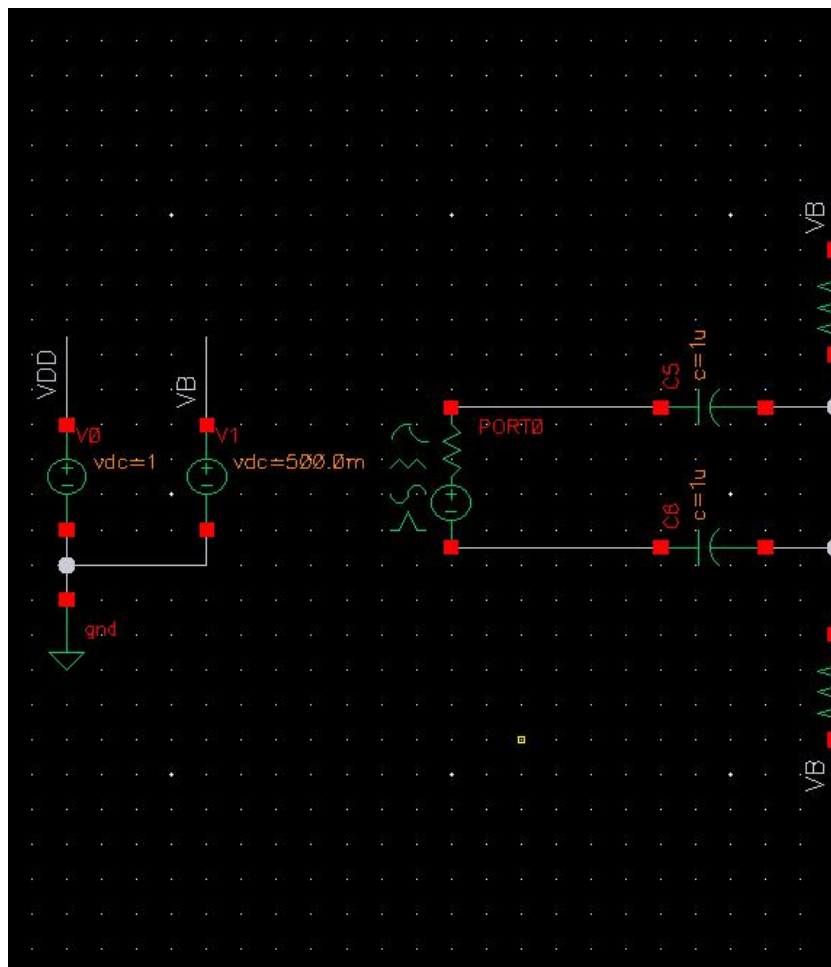
Port mode	Value	Display
Resistance	Real_ZS Ohms	off
Reactance	Imag_ZS Ohms	off
Port number		off
DC voltage		off
Source type	sine	off
Frequency name 1	fin	off
Frequency 1	fin Hz	off
Amplitude 1 (Vpk)		off
Amplitude 1 (dBm)	p1n	off
Phase for Sinusoid 1		off
Sine DC level		off
Delay time		off
Display second sinusoid	☐	off
Display multi sinusoid	☐	off
Display modulation params	☐	off
Display small signal params	☐	off
Display temperature params	☐	off
Display noise parameters	☐	off

OK Cancel Apply Defaults Previous Next Help



Testbench搭建

- Testbench中各元件的详细信息



Apply To:

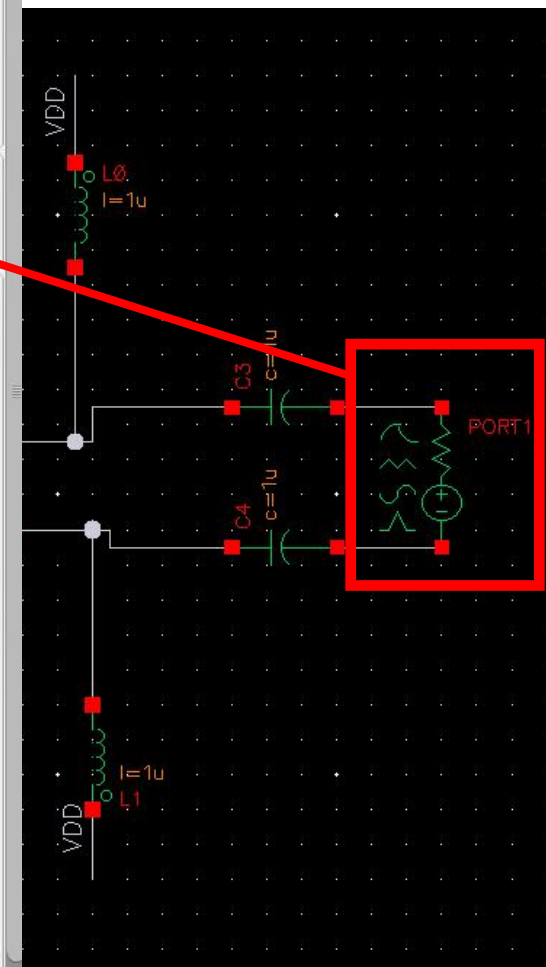
Show: ☐ system ☒ user ☒ CDF

Property	Value	Display
Library Name	analogLib	off
Cell Name	port	off
View Name	symbol	off
Instance Name	PORT1	off

User Property	Master Value	Local Value	Display
lvignore	TRUE		<input checked="" type="button" value="off"/>

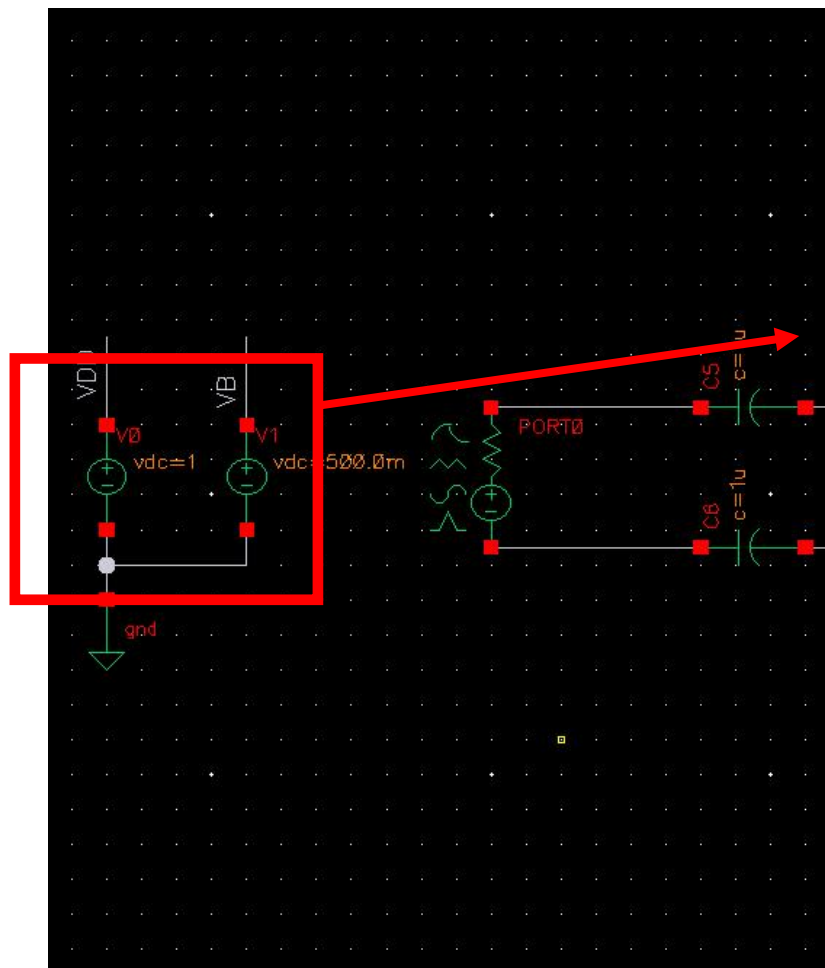
CDF Parameter

Property	Value	Display
Port mode	☒ Normal ☐ HarmonicPort	off
Resistance	Real_ZL Ohms	off
Reactance	Imag_ZL Ohms	off
Port number		off
DC voltage		off
Source type	sine	off
Frequency name 1		off
Frequency 1		off
Amplitude 1 (Vpk)		off
Amplitude 1 (dBm)		off
Phase for Sinusoid 1		off
Sine DC level		off
Delay time		off
Display second sinusoid	☐	off
Display multi sinusoid	☐	off
Display modulation params	☐	off
Display small signal params	☐	off
Display temperature params	☐	off
Display noise parameters	☐	off



Testbench搭建

- Testbench中各元件的详细信息



Apply To: ☒ only current ☒ instance

Show: ☐ system ☒ user ☒ CDF

Browse Reset Instance Labels Display

Property	Value	Display
Library Name	analogLib	off
Cell Name	vdc	off
View Name	symbol	off
Instance Name	V0	off

Add Delete Modify

User Property	Master Value	Local Value	Display
Ivsignore	TRUE		off

CDF Parameter

Property	Value	Display
Noise file name		off
Number of noise/freq pairs	0	off
DC voltage	1 V	off
AC magnitude		off
AC phase		off
XF magnitude		off
PAC magnitude		off
PAC phase		off
Temperature coefficient 1		off
Temperature coefficient 2		off
Nominal temperature		off

OK Cancel Apply Defaults Previous Next Help

Apply To: ☒ only current ☒ instance

Show: ☐ system ☒ user ☒ CDF

Browse Reset Instance Labels Display

Property	Value	Display
Library Name	analogLib	off
Cell Name	vdc	off
View Name	symbol	off
Instance Name	V1	off

Add Delete Modify

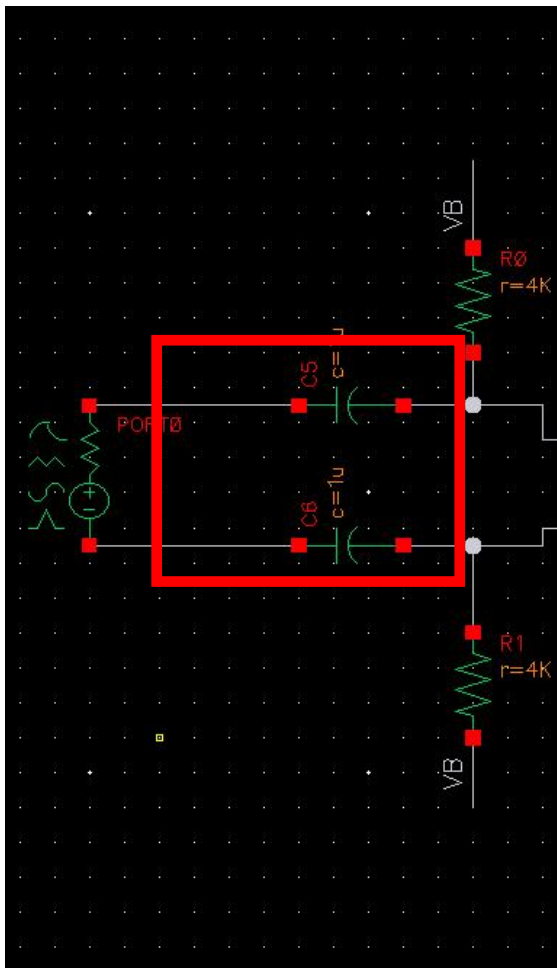
User Property	Master Value	Local Value	Display
Ivsignore	TRUE		off

CDF Parameter

Property	Value	Display
Noise file name		off
Number of noise/freq pairs	0	off
DC voltage	500.0m V	off
AC magnitude		off
AC phase		off
XF magnitude		off
PAC magnitude		off
PAC phase		off
Temperature coefficient 1		off
Temperature coefficient 2		off
Nominal temperature		off

OK Cancel Apply Defaults Previous Next Help

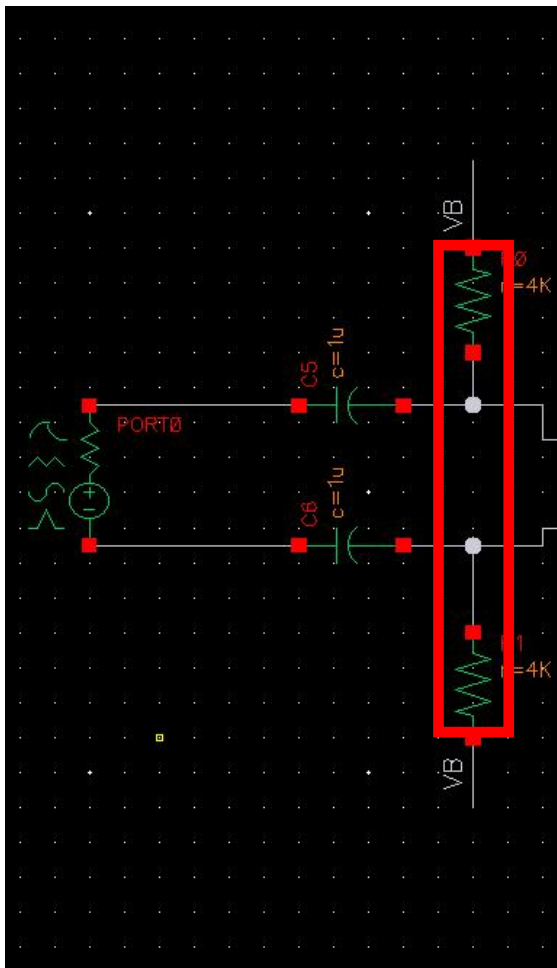
- Testbench中各元件解释



输入串联电容的主要作用是隔离直流电压，同时让交流信号通过。输入信号源 PORT0提供纯交流信号，而差分放大器的输入端（晶体管的栅极）需要一个精确的直流偏置电压VB才能正常工作在放大区。串联电容将这两者隔离开；输出同理。

Testbench搭建

- Testbench中各元件解释



4k Ω 电阻:

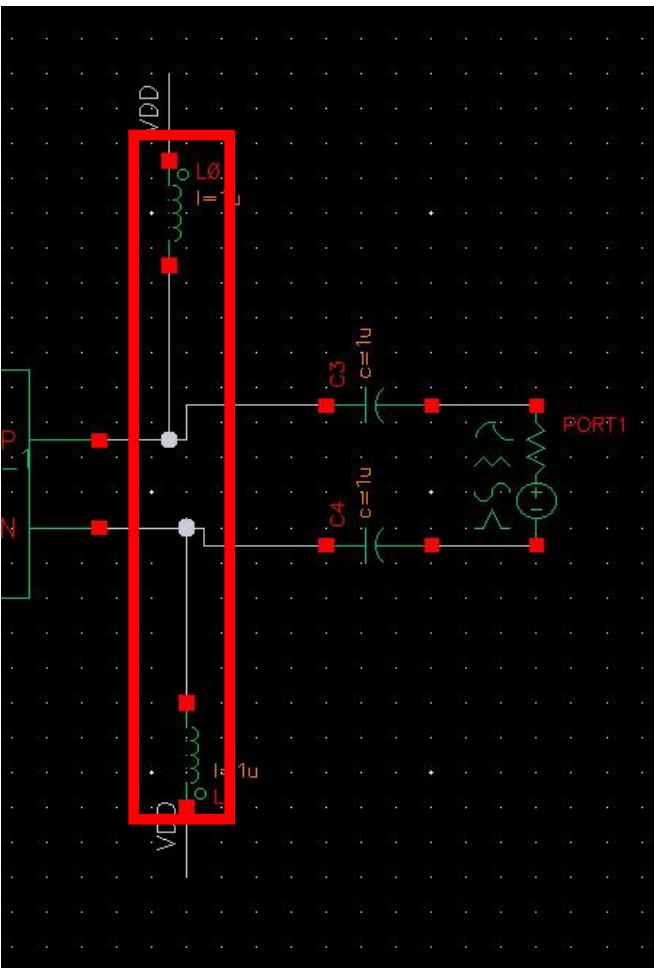
提供栅极偏置。

这两个电阻为差分对中的 NMOS 晶体管的栅极提供一个稳定的直流工作电压。电压源 V1 产生一个 VB 直流电压，通过 R0 和 R1 分别加载到 VINP 和 VINN 节点。

选择 4k Ω 的电阻值是为了在提供稳定偏置的同时，对输入信号源呈现足够大的输入阻抗，避免过度消耗输入信号的能量。

Testbench搭建

- Testbench中各元件解释



输出并联电感：
提供RF Choke。

对直流电而言，电感的阻抗几乎为零，因此它们可以高效地将电源电压 VDD 提供的直流电流输送给 NMOS 晶体管的漏极，而不会产生像电阻负载那样的直流电压降。

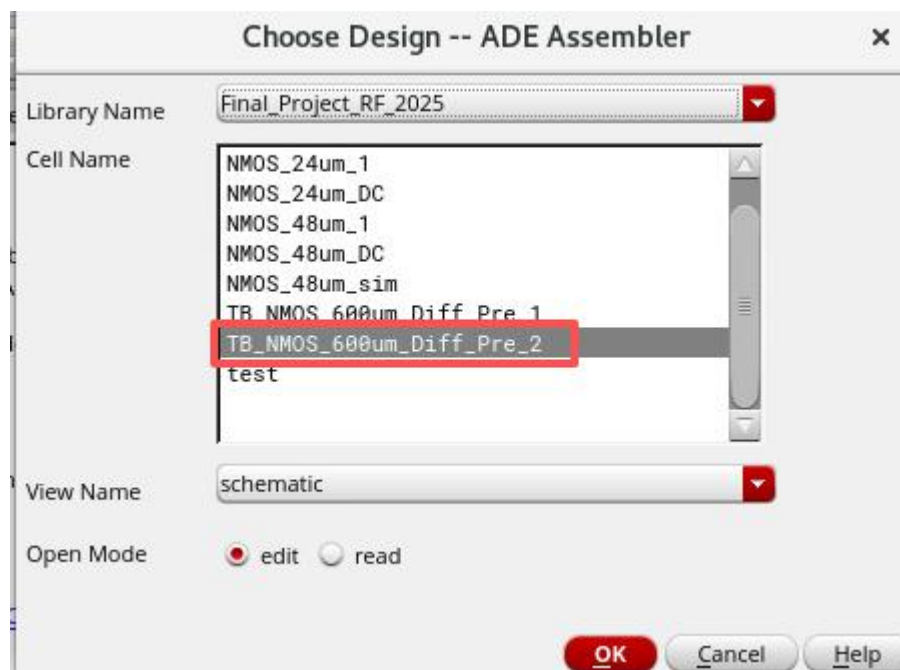
Testbench搭建

- 仿真环境设置
- 按照《cadence virtuoso 使用教程》中的步骤launch ADE XL或ADE Assembler，如图所示：



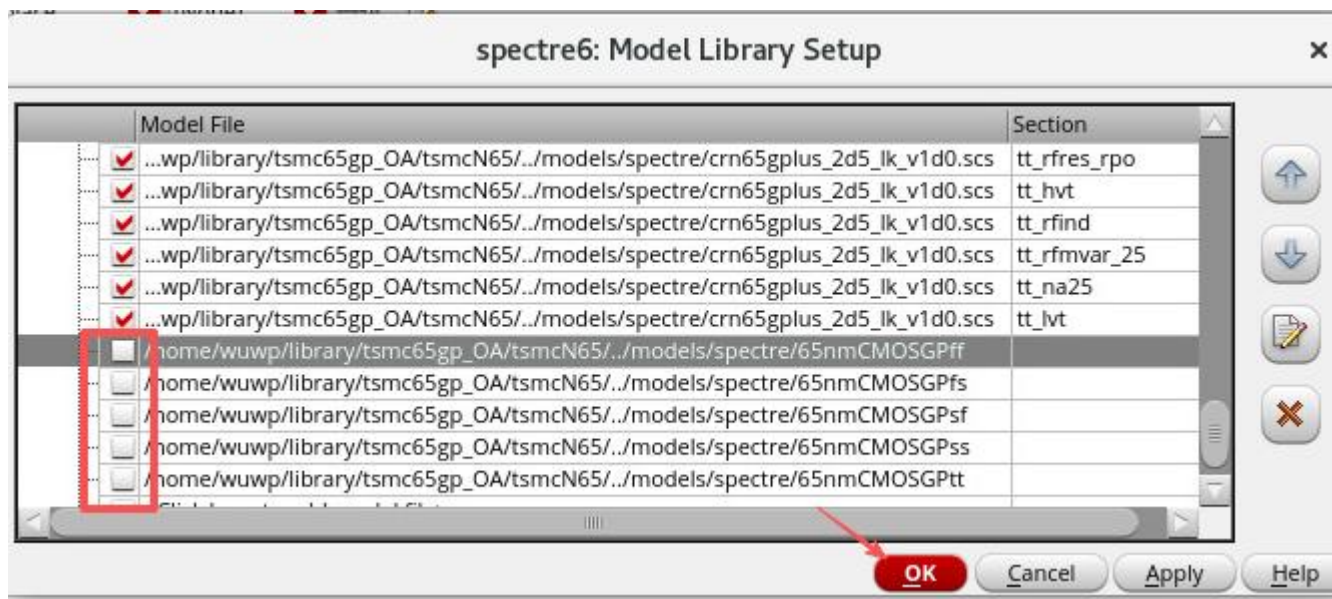
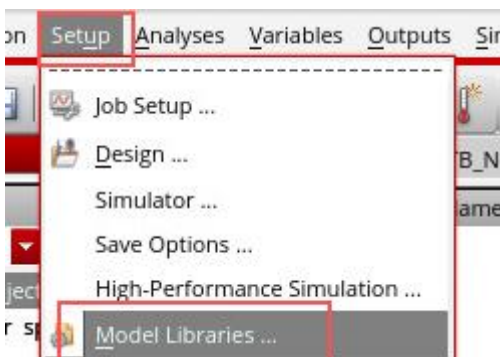
Testbench搭建

- 点击click to add test, 将当前的schematic关联至仿真文件



Testbench搭建

- 点击菜单栏Setup→Model Libraries
- 取消勾选library setup的后五个文件，OK



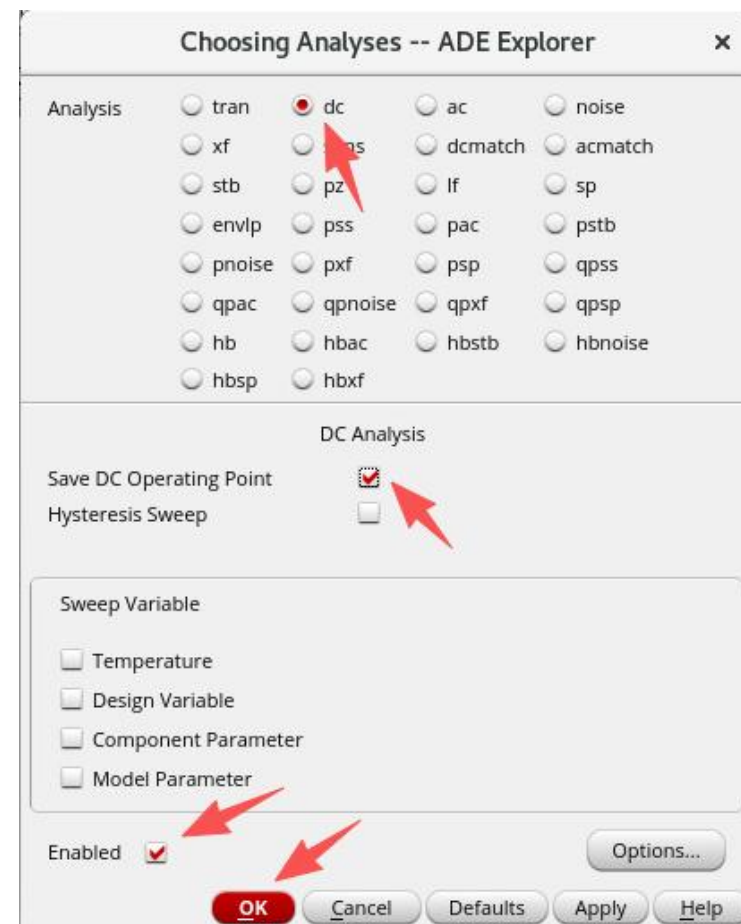
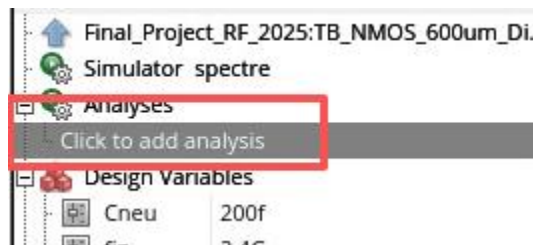
Testbench搭建

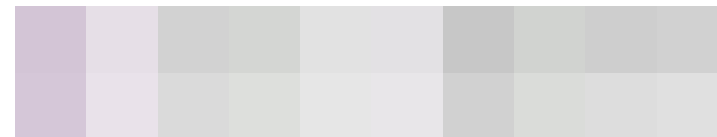
- 将testbench中的design variable加载到仿真环境
- 点击“Design Variable” 左侧加号
- 右键“Design Variable”， 点击copy from cellview
- Schematic中的设计变量出现在仿真环境中
- 给各变量分别赋初始值， 如图所示



Testbench搭建

- 添加仿真Analyses
- 点击“Analyses”左侧加号 → click to add analysis
- 首先选择DC仿真，如图所示
- 无论进行何种类型的仿真，请养成先跑DC的习惯，便于帮助大家检查DC工作点

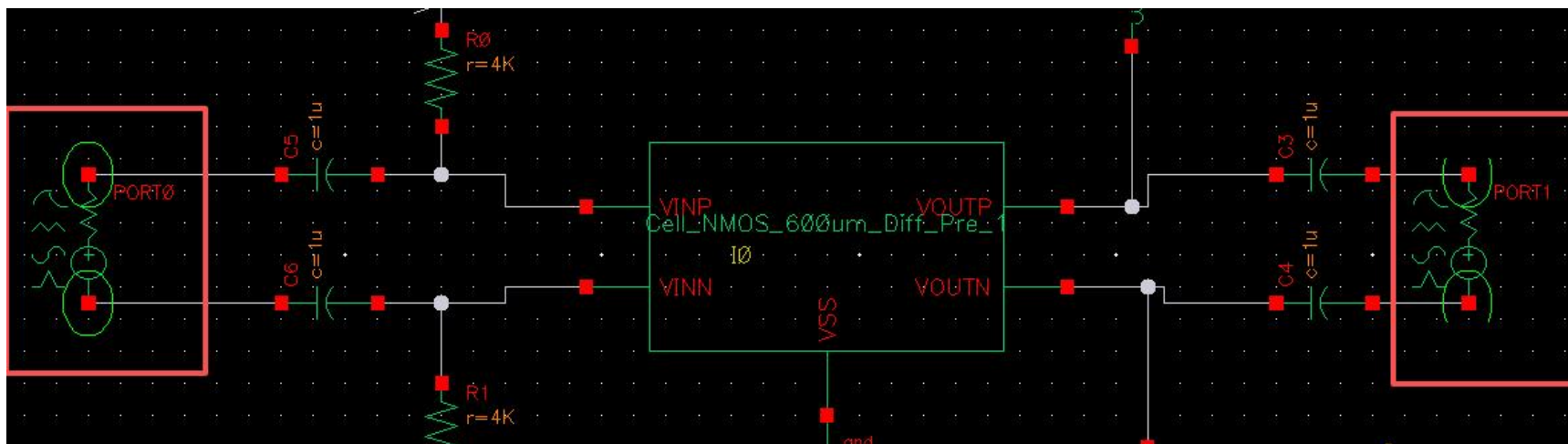
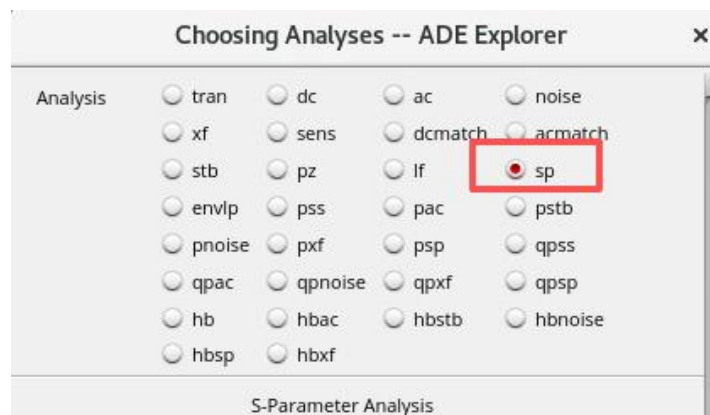




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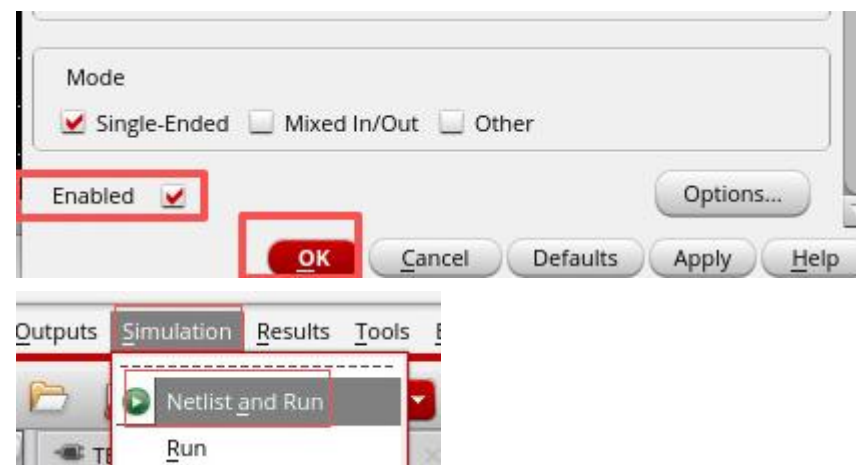
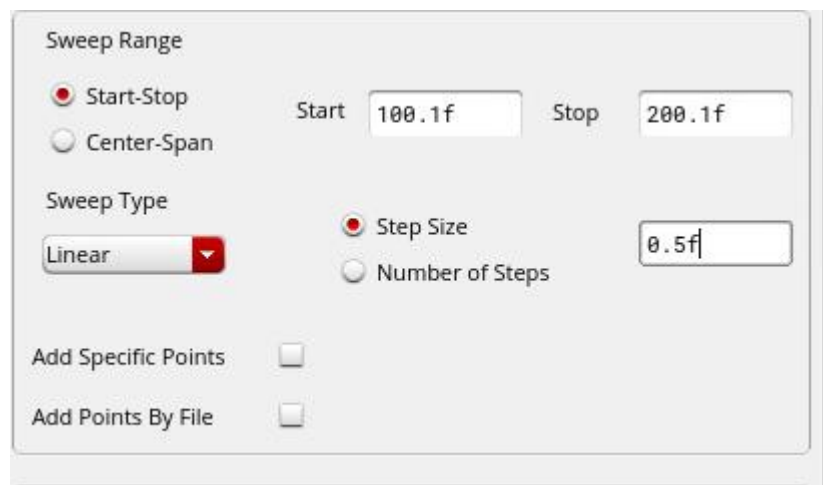
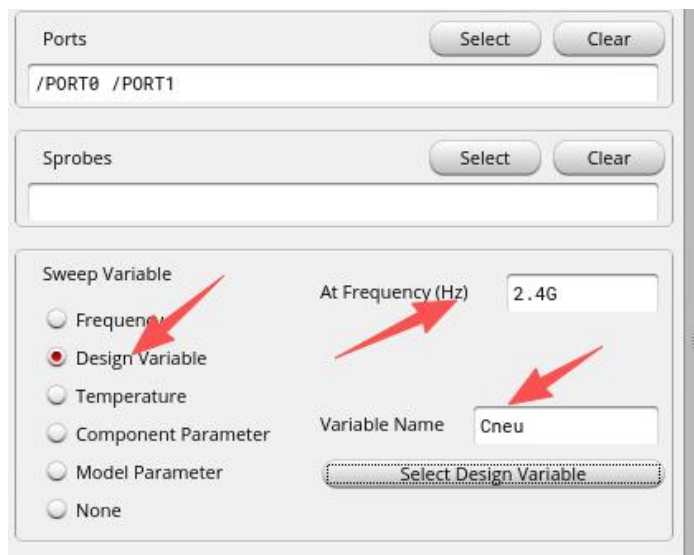
中和电容选取

- 中和电容选取的准则是使得Kf最优
- 继续点击Analyses下方的click to add analysis, 选择SP仿真
- 在Ports选择框右上方点击Select, 自动回到testbench, 分别点击输入输出port, 然后按ESC完成port选择



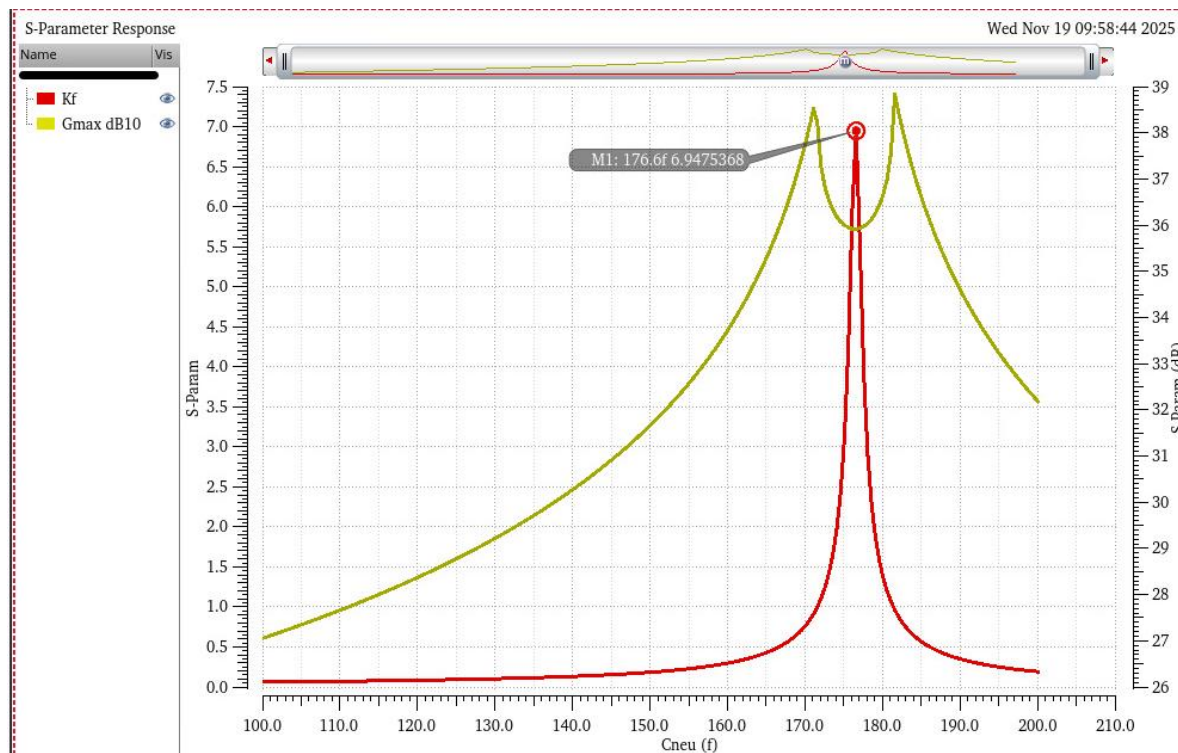
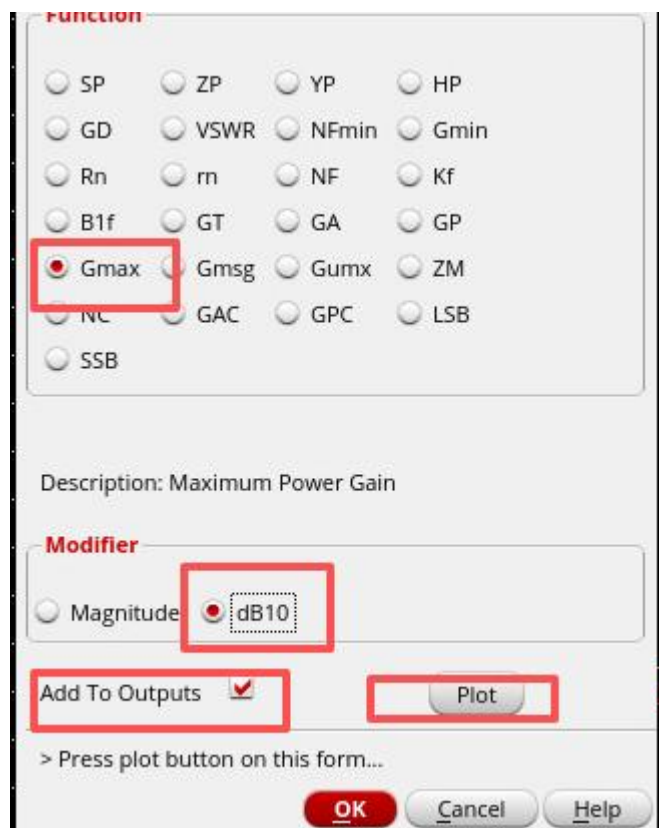
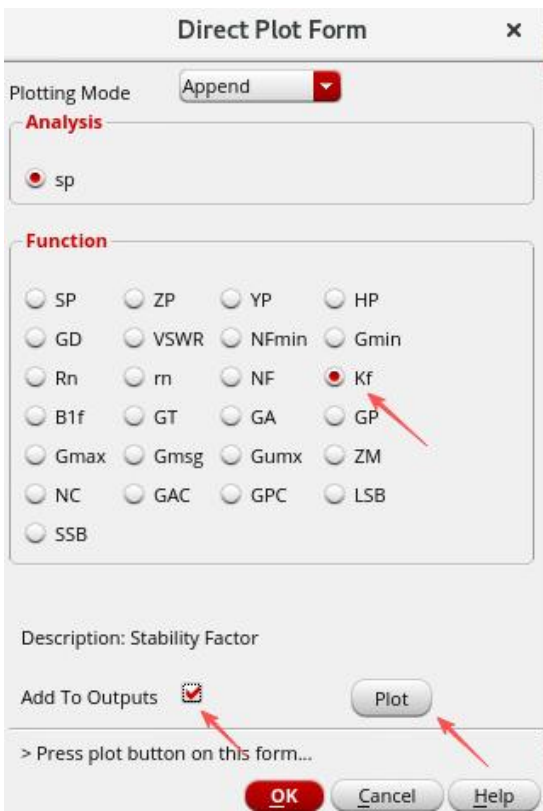
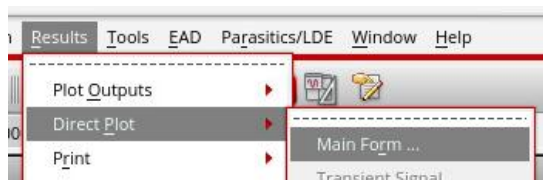
中和电容选取

- 在Sweep variable处点击Design Variable
- At Frequency处填写设计的中心频点2.4G
- Variable Name处选择Select Design Variable→Cneu
- 在Sweep Range处设置Cneu的扫描范围和步长，如图所示
- 随后点击OK，并在仿真环境中run simulation



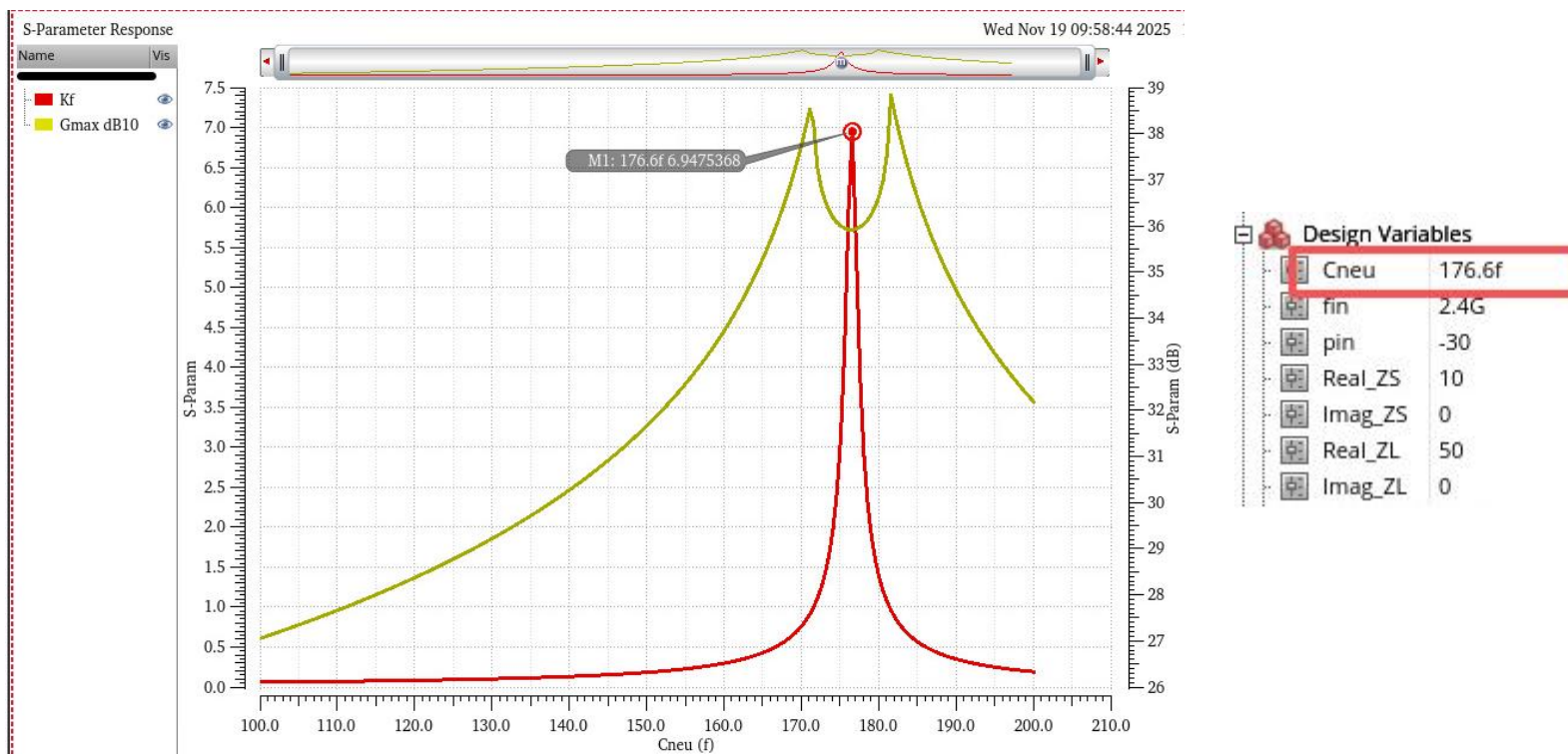
中和电容选取

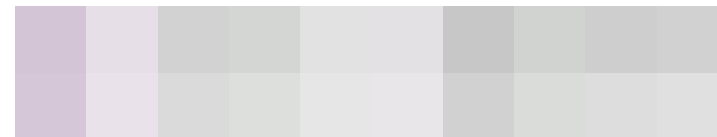
- Simulation完成后，点击Results → Direct Plot → Main Form
- 在direct plot form窗口依次选择Kf和Gmax，如图所示



中和电容选取

- 由Kf vs Cneu曲线可以得知，Cneu=176.6fF时，Kf最优，此时Gmax处于最低点
- 回到仿真环境，在Design Variable中将Cneu的值固定为176.6fF

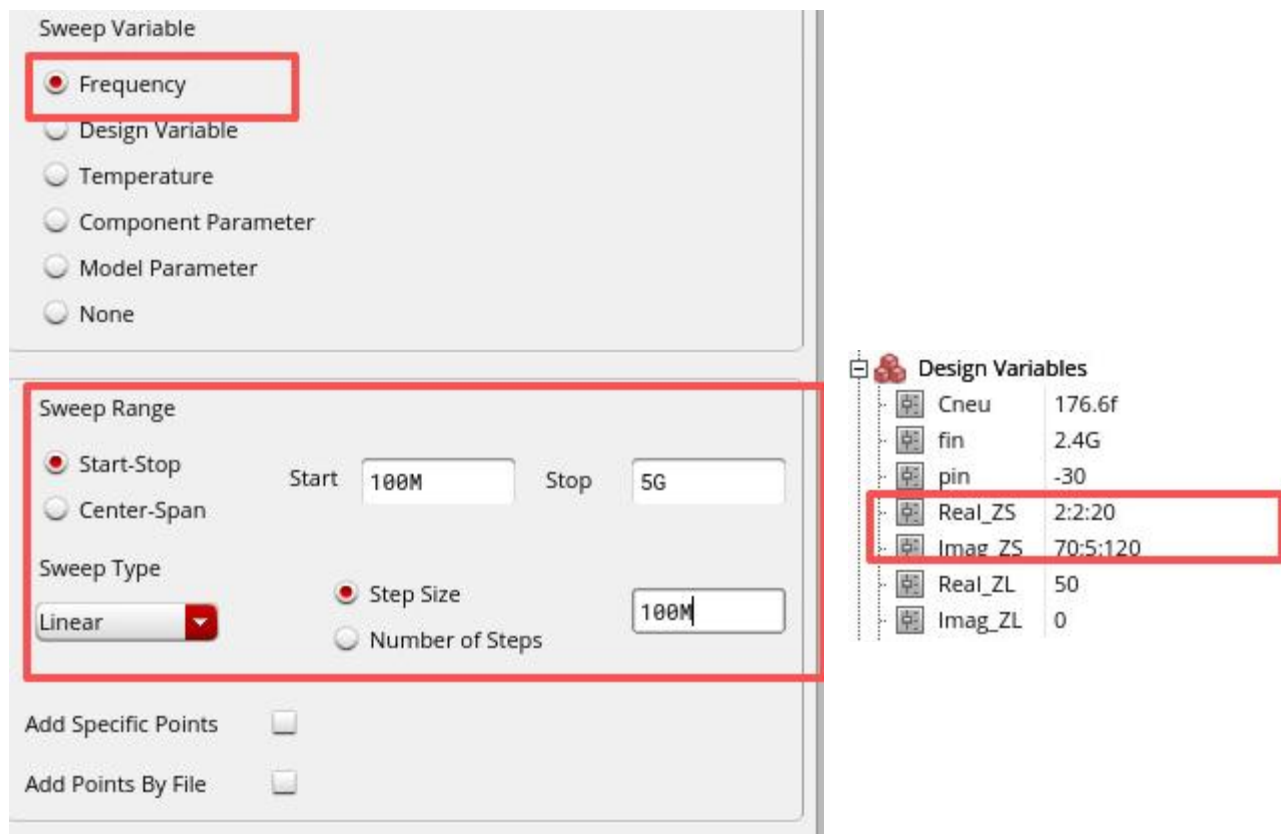




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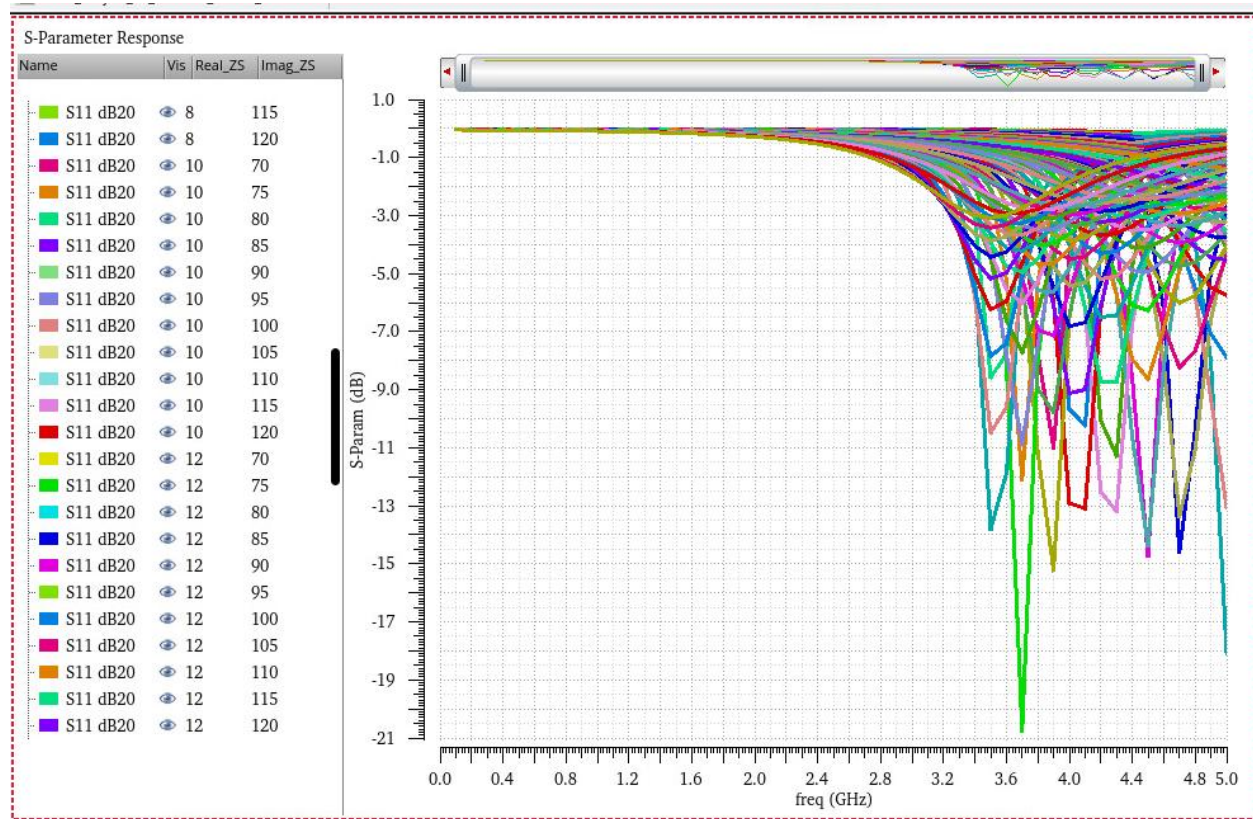
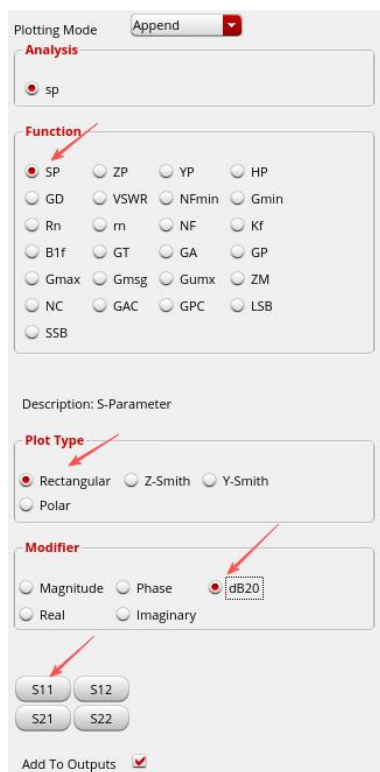
小信号仿真确认输入阻抗

- 输入阻抗确认的准则是输入回波损耗趋近于无限低
- 双击sp仿真，将扫描变量从Cneu修改为频率
- 在Design Variable中扫描阻抗ZS的实部和虚部，扫描范围自行定义，注意控制总扫描点不要过多，如图所示
- 运行仿真



小信号仿真确认输入阻抗

- Simulation完成后，点击Results → Direct Plot → Main Form
- 在direct plot form窗口选择S11 (dB)，如图所示
- 观察曲线趋势，修正扫描范围

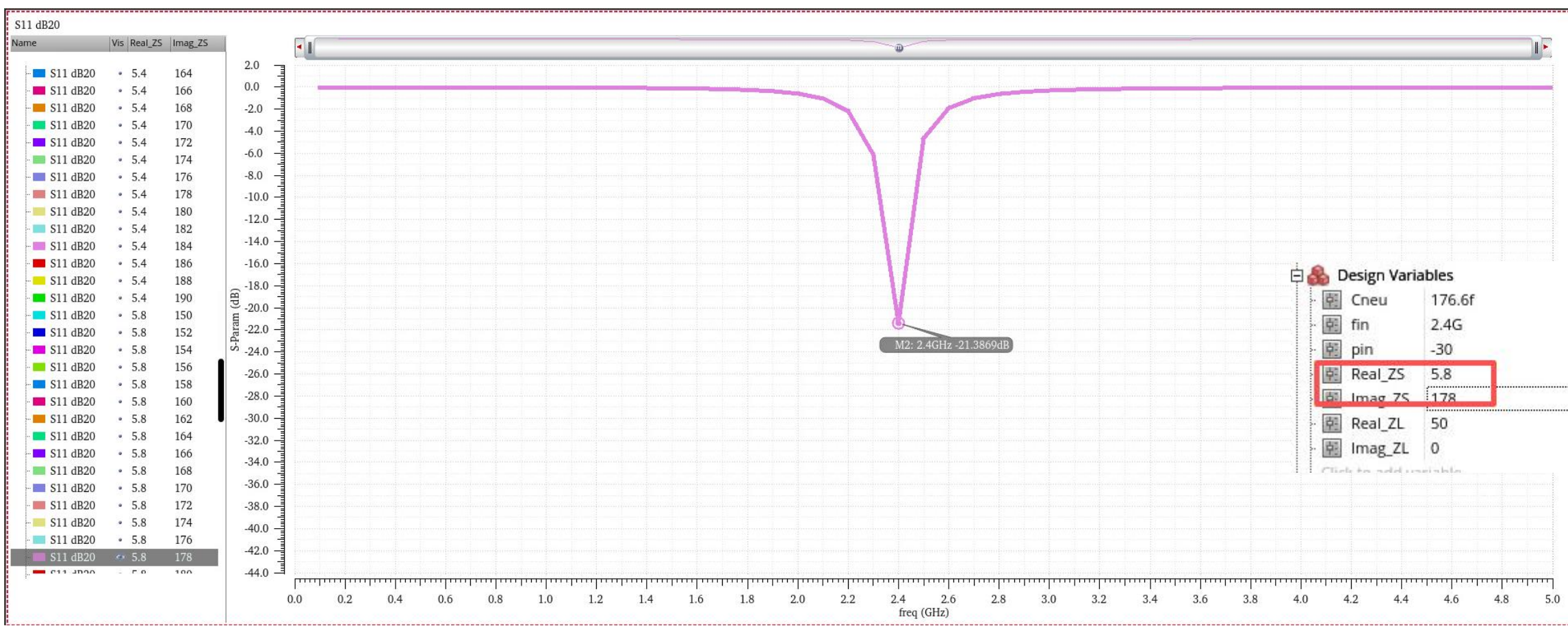


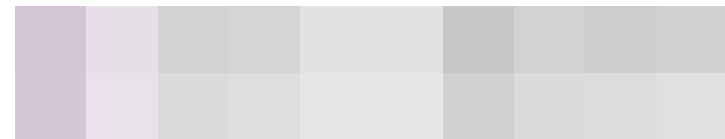
Design Variables

Cneu	176.6f
fin	2.4G
pin	-30
Real_ZS	3:0.4:7
Imag_ZS	150:2:190
Real_ZL	50
Imag_ZL	0

小信号仿真确认输入阻抗

- 再次观察S11的趋势，确定一个能够使S11在2.4GHz足够小(<-20dB)的阻抗值，更新到Design Variable中

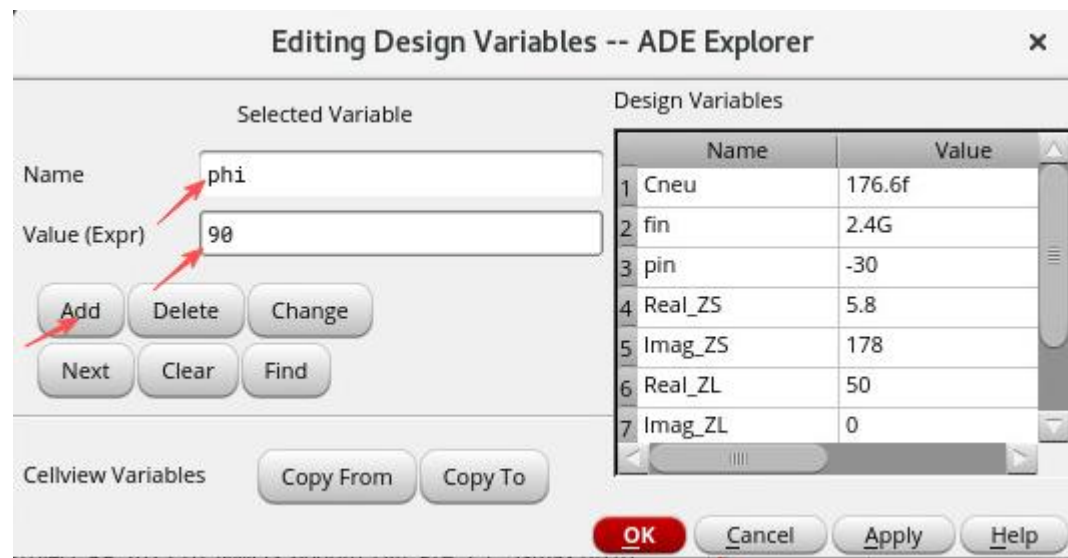
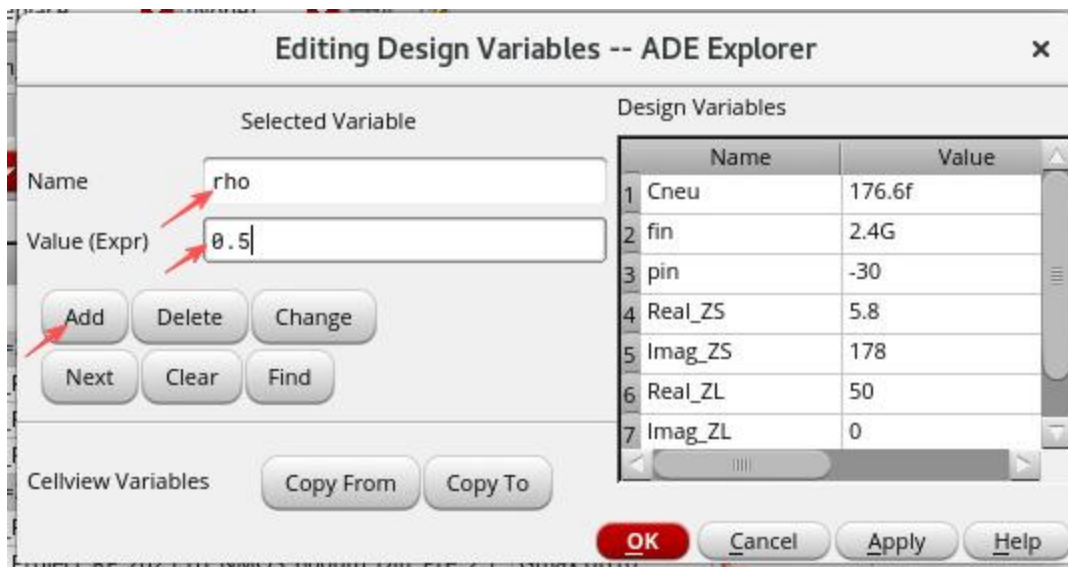




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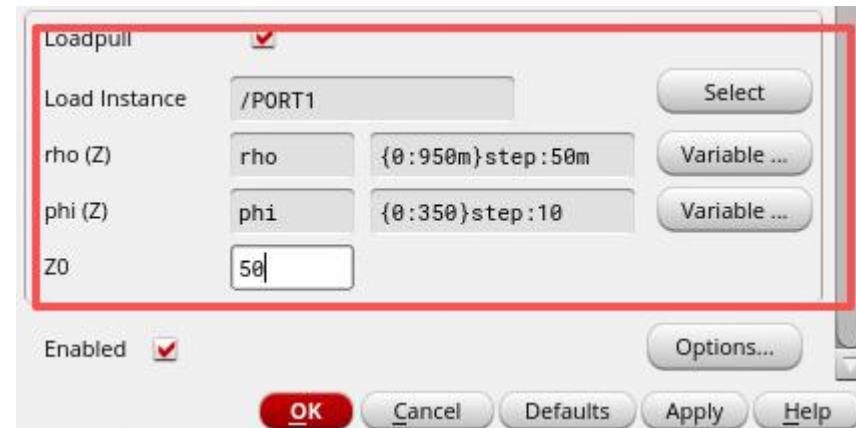
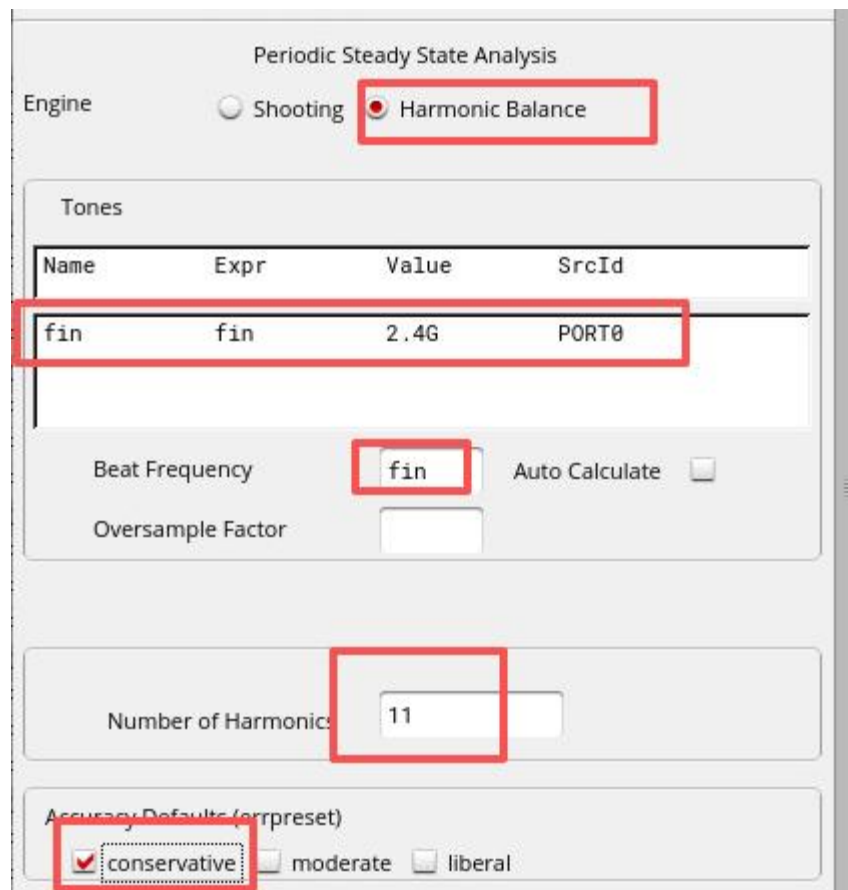
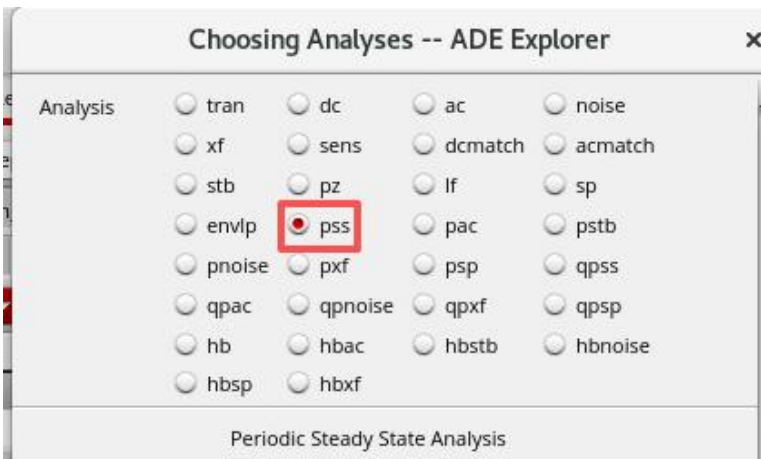
Loadpull仿真确认输出阻抗

- 在设置Loadpull仿真之前，首先在Design Variable中添加两个变量，分别为幅度（rho）和相位（phi），如图所示
- 这两个变量用于仿真器进行loadpull时的扫描，因此此处初始值可无需特别设计



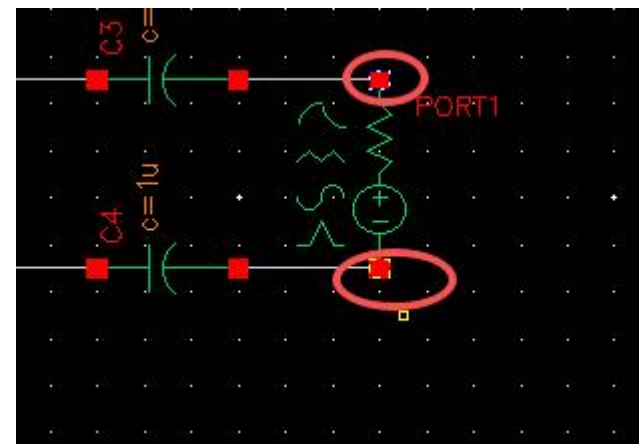
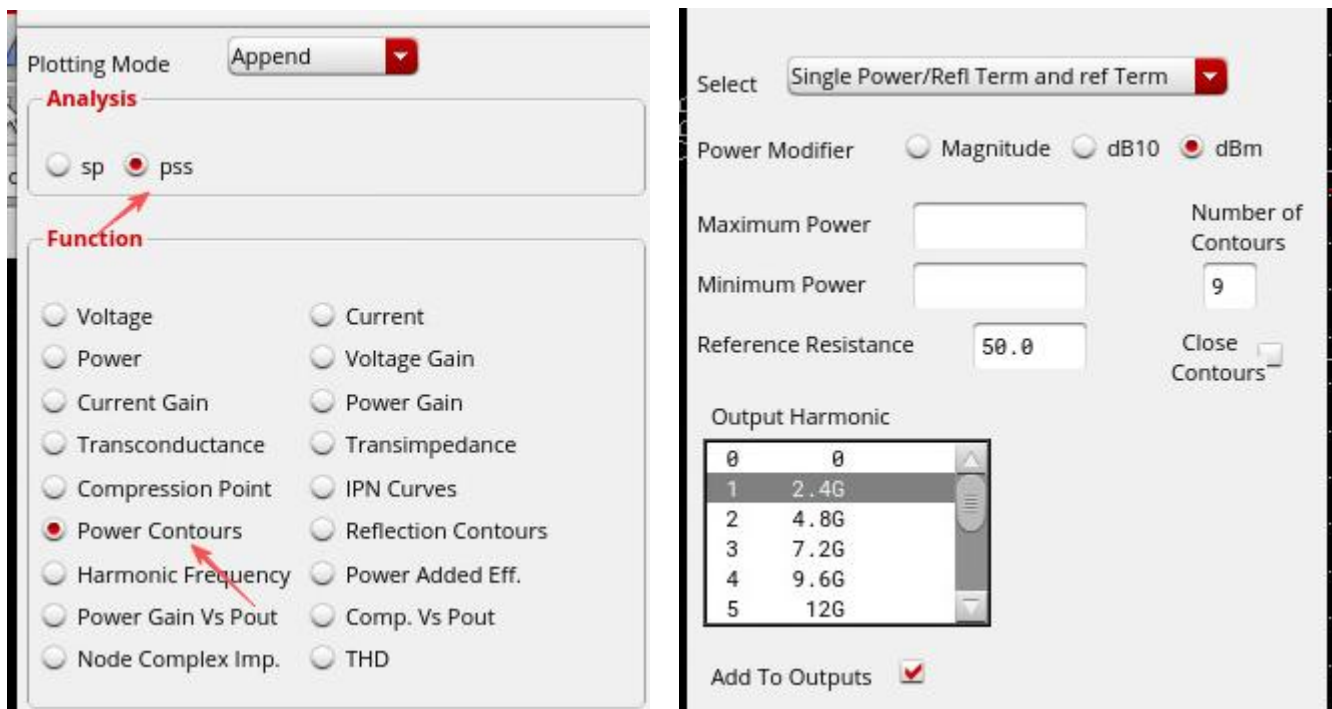
Loadpull仿真确认输出阻抗

- 点击Analyses → click to add analysis, 添加pss仿真
- 选择pss仿真中的hb (Harmonic Balance)仿真, 设置参数如下
- 勾选Loadpull, 选择Load Instance为输出port, 在rho和phi框中分别设置幅度和相位扫描范围, 运行仿真



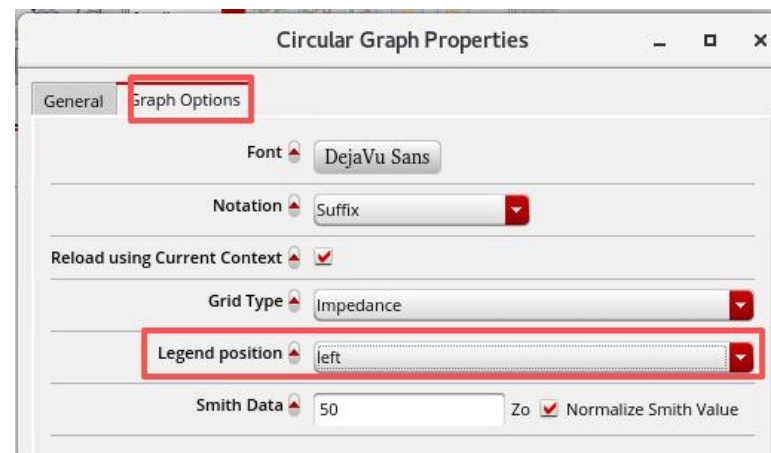
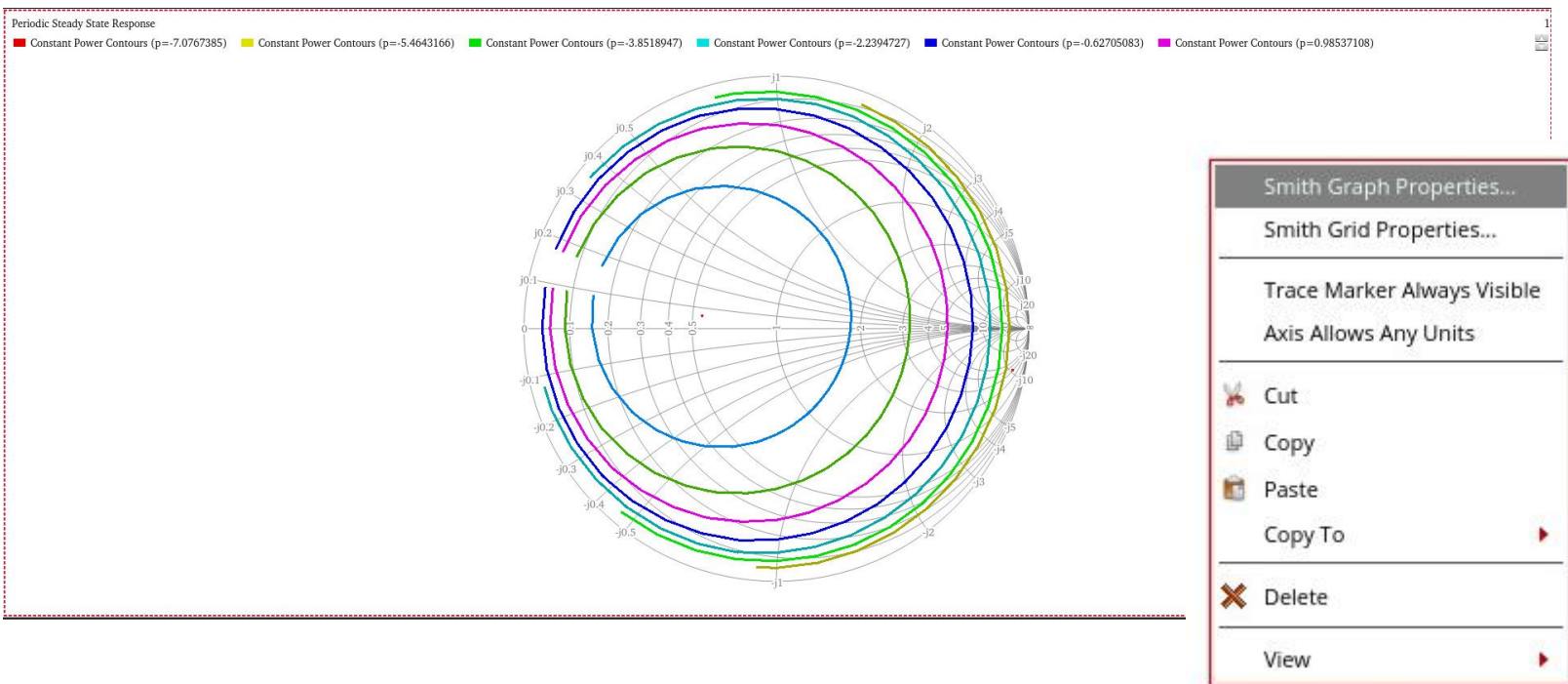
Loadpull仿真确认输出阻抗

- 仿真完成后，点击Results → Direct Plot → Main Form
- 在direct plot form窗口选择pss→Power Contours，详细设置如图所示
- 设置好后，不要点击OK，依次点击原理图中输出port的正负terminal



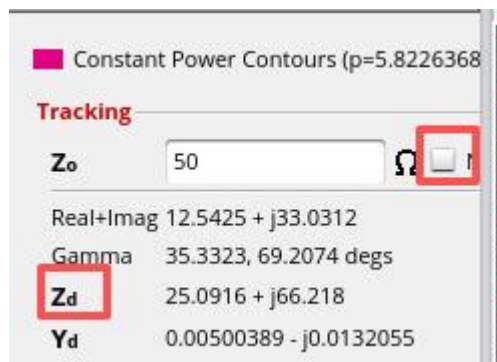
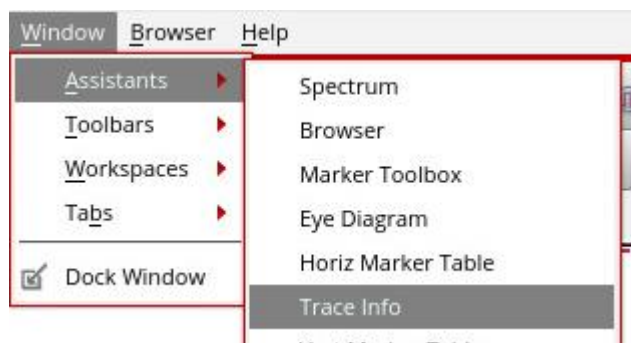
Loadpull仿真确认输出阻抗

- 等功率圆如图所示。
- 右键smith graph properties, 可以将legend position设为left, 方便观察功率值



Loadpull仿真确认输出阻抗

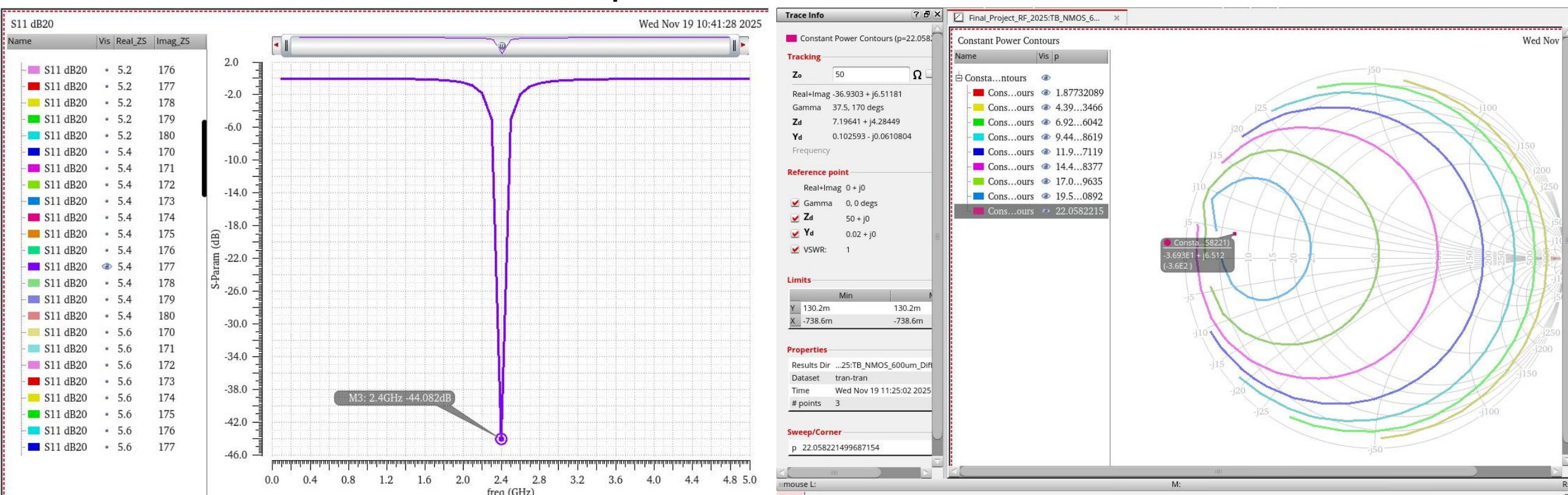
- 点击菜单栏window-assistants-trace info, 可显示鼠标位置的阻抗值
- 取消勾选normalize, 此时Zd跟随鼠标显示实际阻抗值
- 将鼠标停靠在等功率圆图的最优功率点上, 记录此时的Zd值, 并更新到仿真环境 Design Variable

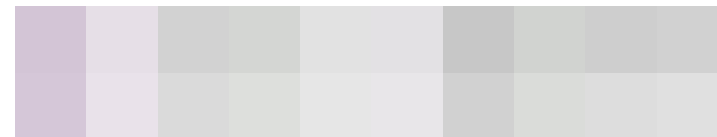


Real_ZL	7.19
Imag_ZL	4.28

Loadpull仿真确认输出阻抗

- 由于变化输出阻抗会影响输入阻抗，因此需微调ZS的实部和虚部，根据S11确定新的输入阻抗值
- 在新的ZS下进行第二次Loadpull，确定新的输出阻抗

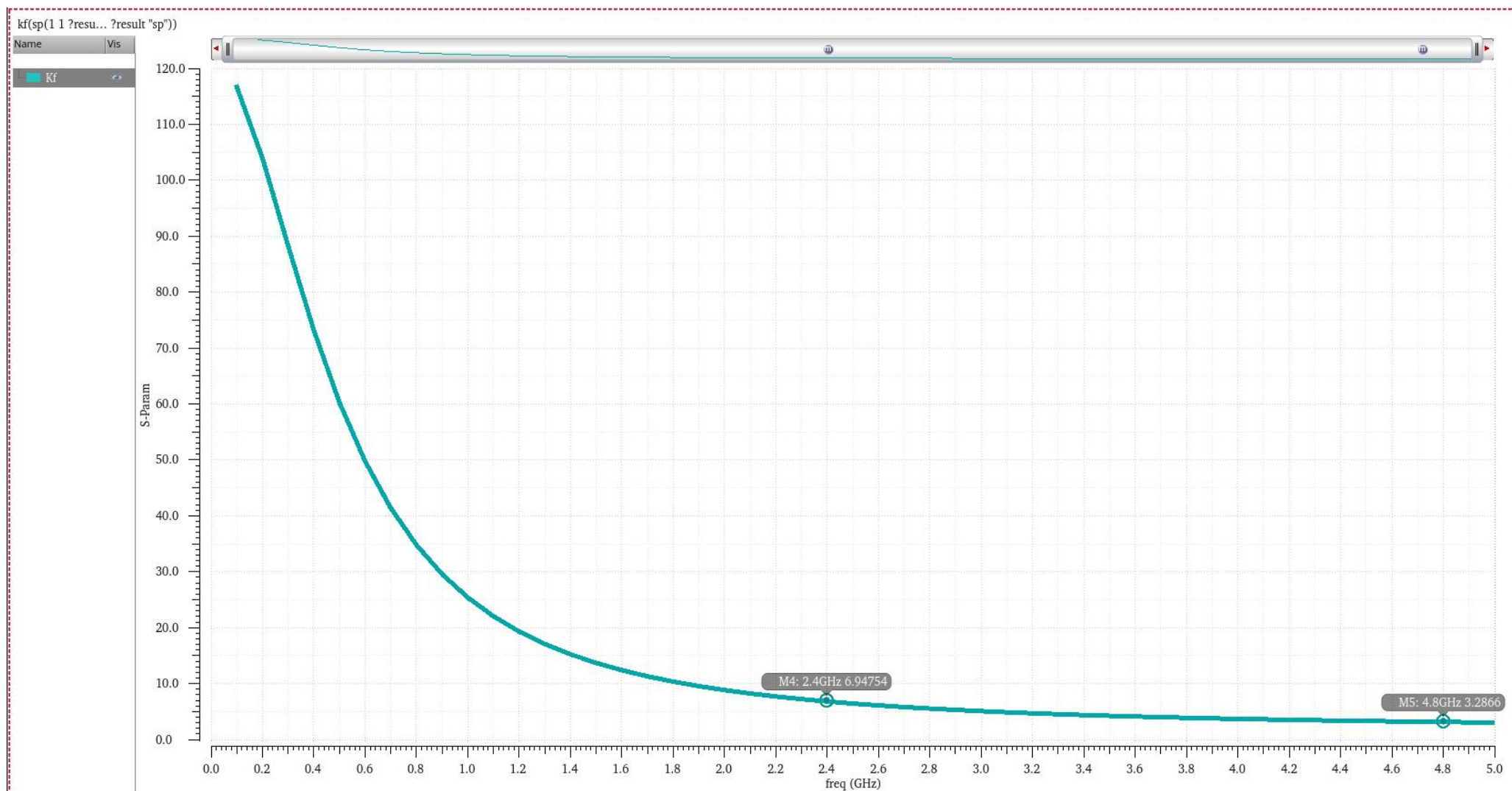




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- Loadpull仿真确认输出阻抗
- 核心性能获取

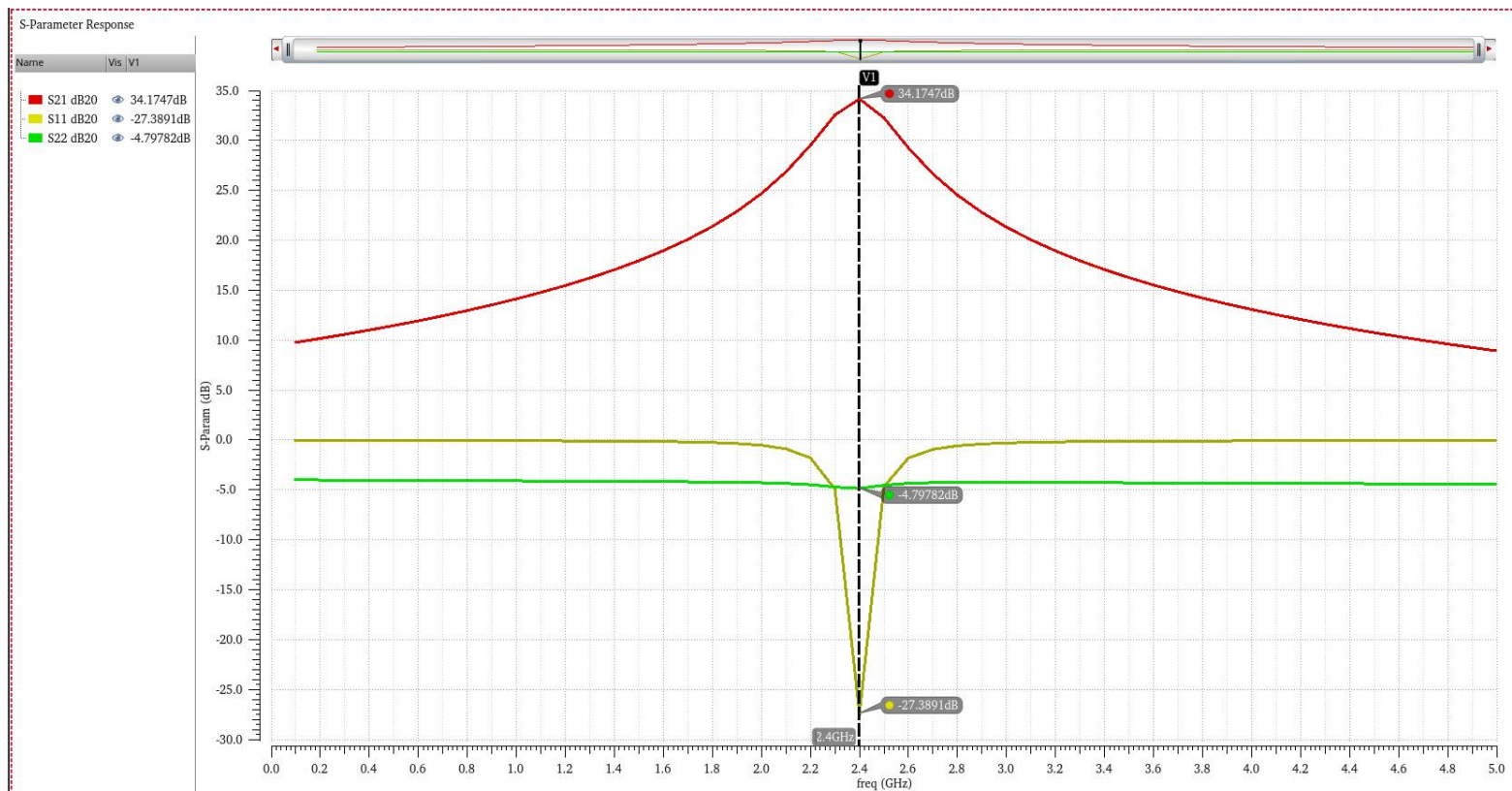
核心性能获取

- 小信号：Kf



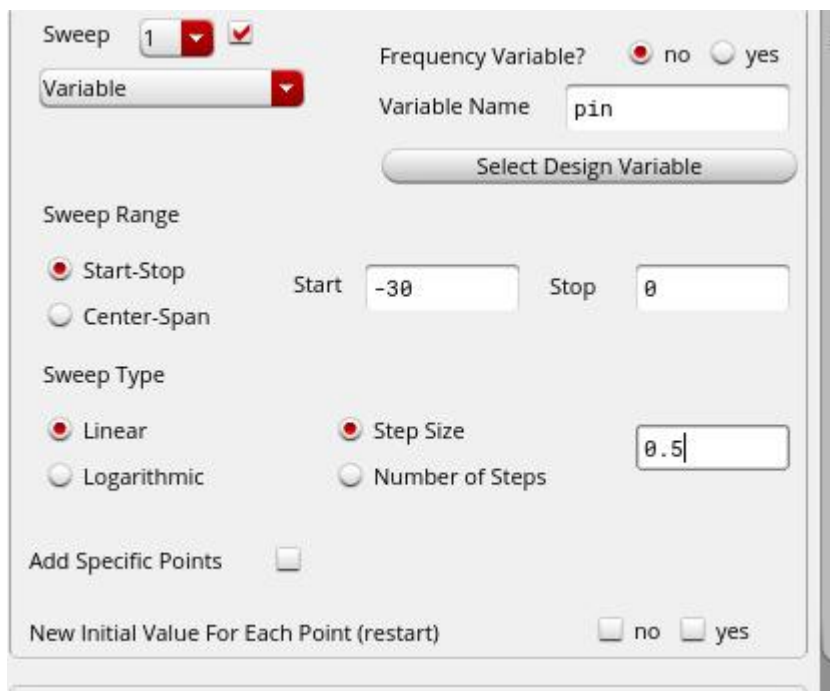
核心性能获取

- 小信号：S11 S22 S21
- 注意，由于输出阻抗是大信号匹配，因此S22不是最优情况



核心性能获取

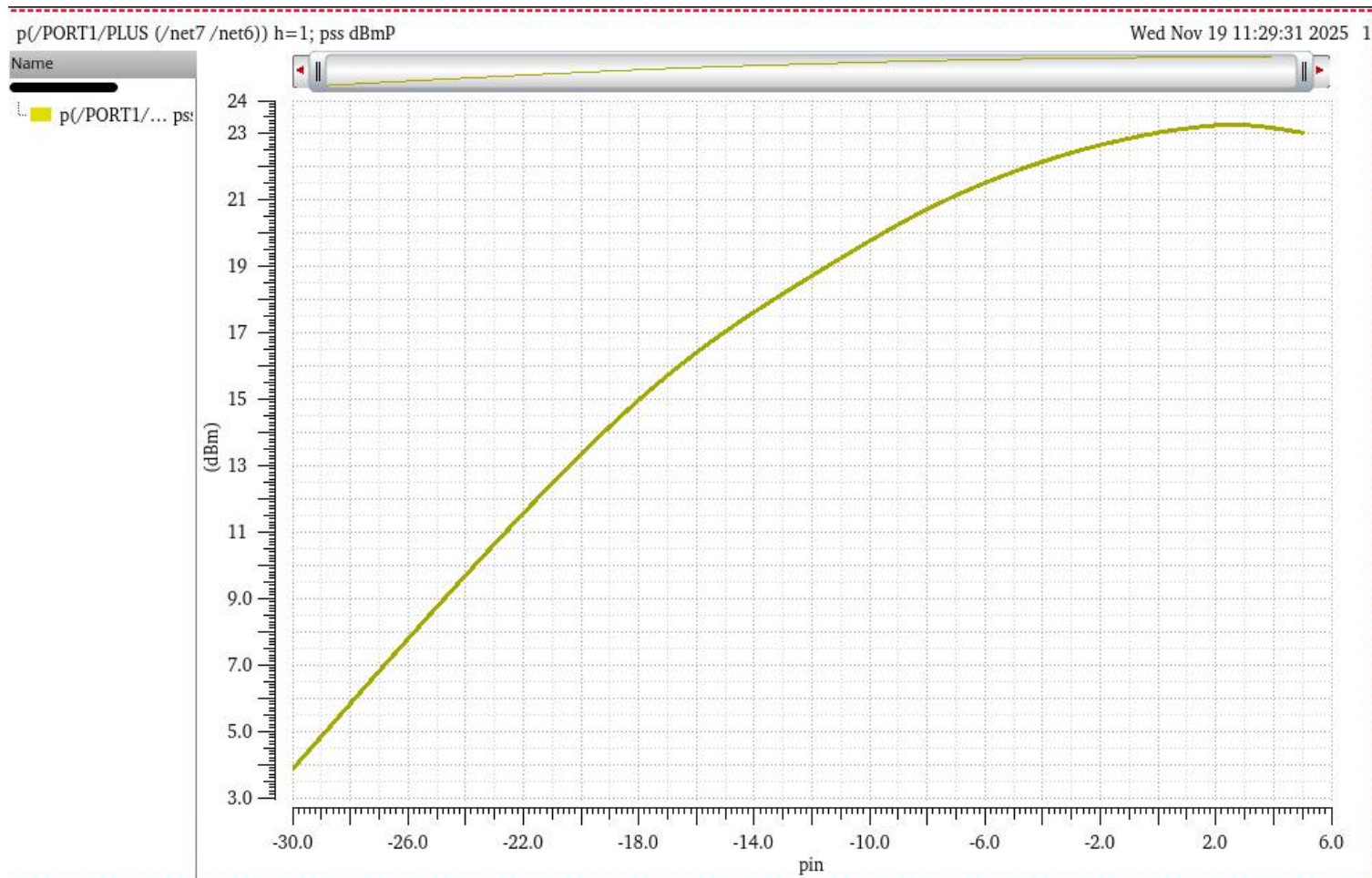
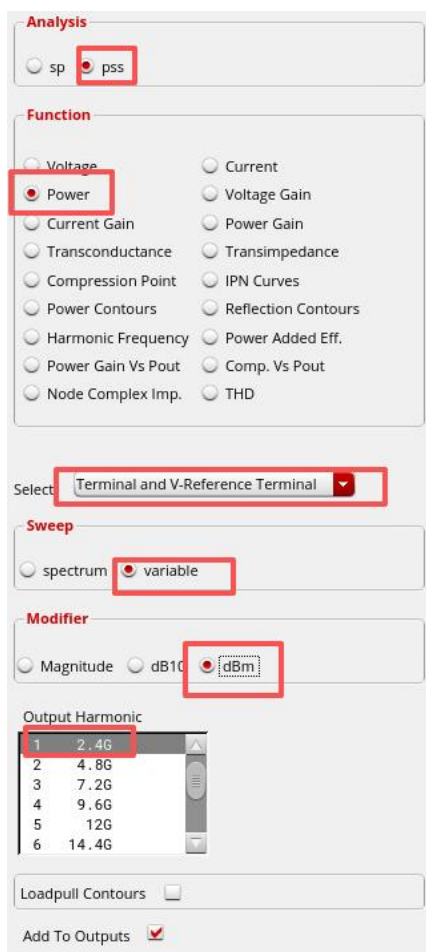
- 大信号：输出功率曲线
- 在analyses中设置pss，取消勾选loadpull，选择sweep，设置细节如下图所示



The screenshot shows a 'Sweep' configuration dialog box. At the top, 'Sweep' is set to '1' with a red checkmark. Below it, 'Variable' is selected from a dropdown menu. To the right, 'Frequency Variable?' has 'no' selected with a red radio button. 'Variable Name' is set to 'pin'. A 'Select Design Variable' button is below this. The 'Sweep Range' section has 'Start-Stop' selected with a red radio button, with 'Start' at -30 and 'Stop' at 0. The 'Sweep Type' section has 'Linear' selected with a red radio button. 'Step Size' is set to 0.5, and 'Number of Steps' is unselected. 'Add Specific Points' is unselected. At the bottom, 'New Initial Value For Each Point (restart)' has 'no' selected.

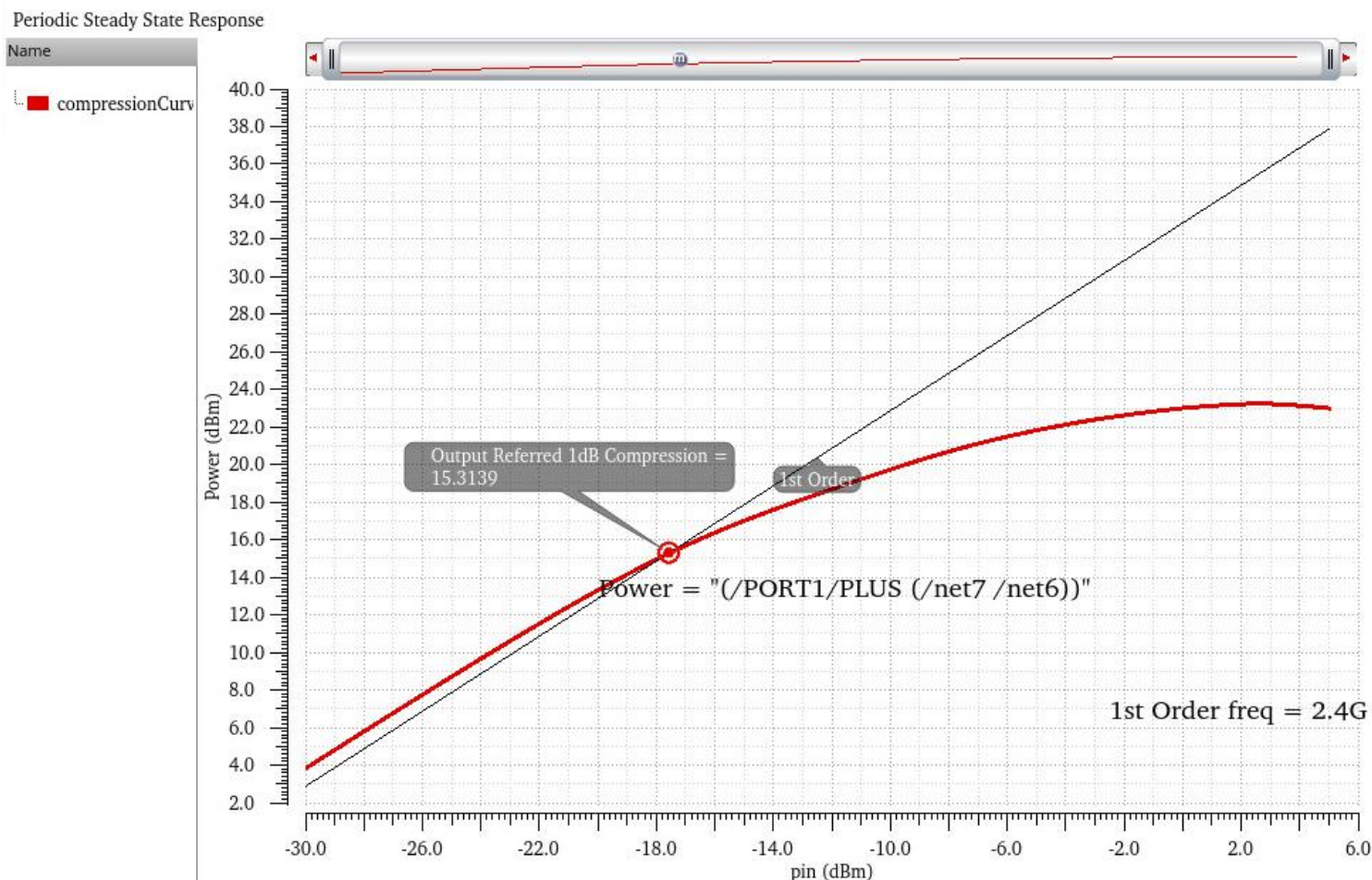
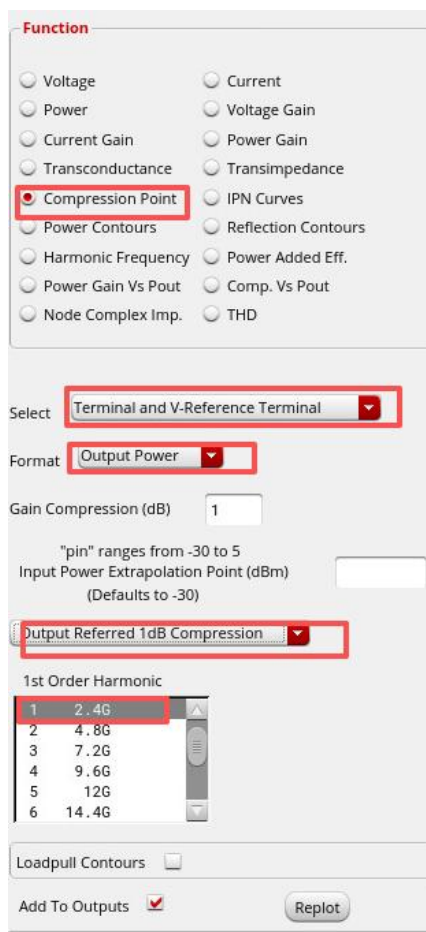
核心性能获取

- 仿真完成后，点击result-direct plot-main form
- 设置细节如下图所示，设置好后先不要点OK，依次点击输出port的正负terminal



核心性能获取

- OP1dB
- 设置细节如下图所示，设置好后先不要点OK，依次点击输出port的正负terminal



核心性能获取

- PAE
- 设置细节如下图所示，设置好后先不要点OK，依次点击输出port的正负terminal、输入port的正负terminal、VDD电压源的正负terminal

